MILITARY STANDARD

SWITCHES AND ASSOCIATED HARDWARE,
SELECTION AND USE OF

TO ALL HOLDERS OF MIL-STD-1132A:

1. THE FOLLOWING SECTIONS (REVISED AND NEW) ARE TO BE ADDED TO MIL-STD-1132A:

SECTION	DATE	SUPERSEDED SECTION	DATE
105A (Rev)	6 June 1979	105 (pp 105.1 thru 105.24)	23 March 1976
114 (New)	6 June 1979	---	---

2. THE FOLLOWING PAGES OF MIL-STD-1132A HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
iii	23 March 1976	iii	REPRINTED WITHOUT CHANGE
iv	6 June 1979	iv	20 June 1977
1	23 March 1976	1	REPRINTED WITHOUT CHANGE
2	6 June 1979	2	20 June 1977
9	6 June 1979	9	20 June 1977
10	23 March 1976	10	REPRINTED WITHOUT CHANGE
100.1	6 June 1979	100.1	23 March 1976
112.1	23 March 1976	112.1	REPRINTED WITHOUT CHANGE
112.2	6 June 1979	112.2	20 June 1977
112.5	23 March 1976	112.5	REPRINTED WITHOUT CHANGE
112.6	6 June 1979	112.6	23 March 1976
112.17	23 March 1976	112.17	REPRINTED WITHOUT CHANGE
112.18	6 June 1979	112.18	20 June 1977
112.19	6 June 1979	112.19	20 June 1977
112.20	20 June 1977	112.20	REPRINTED WITHOUT CHANGE
112.21	6 June 1979	112.21	20 June 1977
112.22	20 June 1977	112.22	REPRINTED WITHOUT CHANGE
112.27	6 June 1979	112.27	20 June 1977
112.28	23 March 1976	112.28	REPRINTED WITHOUT CHANGE
112.31	23 March 1976	112.31	REPRINTED WITHOUT CHANGE
112.32	6 June 1979	112.32	20 June 1977
112.33	20 June 1977	112.33	REPRINTED WITHOUT CHANGE
112.34	6 June 1979	---	---
112.35	6 June 1979	---	---

3. RETAIN THIS NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS.

4. Holders of MIL-STD-1132A will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Air Force - 85, 17
DLA - ES

User activity:
Navy - CG

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-1081)

FSC 5930

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FOREWORD

This standard provides selected standard switches and associated hardware which are mandatory for use in the design of military equipment.

The application information contained in this standard is offered for guidance only and is not to be considered as mandatory. Additional application information will be added when coordinated with the three military departments.

Additional sections of this standard will be developed as the standard switches and associated hardware of a given specification family are selected and coordinated with the three military departments.

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SECTIONS

Section 100	SWITCHES	100.1
200	ASSOCIATED SWITCH HARDWARE	200.1

CROSS REFERENCE
(Section number to specification number)

Switches

Section		Applicable specification(s)
101	(Reed)	MIL-S-55433
102	(Push button, Illuminated)	MIL-S-22885
103	(Interlock)	MIL-S-8805
104	(Multistation)	MIL-S-24317
105	(Pressure)	MIL-S-9395
106	(Push Button)	MIL-S-8805
107	(Rotary)	MIL-S-3786, MIL-S-6807, and MIL-S-15291
108	(Rotary, Electro-Mechanical Actuated, Solenoid Type)	MIL-S-3786
109	(Rotary, Printed Circuit, Thumbwheel and Pushbutton)	MIL-S-22710
110	(Sensitive)	MIL-S-8805
111	(Thermostatic)	MIL-S-24236 and MIL-S-28827
112	(Toggle)	MIL-S-3950, MIL-S-8834, and MIL-S-8805
113	(Modular Switch Assemblies)	MIL-S-22885
114	(Air and Liquid Flow)	MIL-S-28788

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1. SCOPE

1.1 Scope. This standard establishes requirements for selection of switches and their associated hardware, shown in detail in sections 100 and 200, for use in military equipment. Included are the following:

(a) Selected standard switch types and associated hardware chosen jointly by the Departments of the Army, the Navy, and the Air Force for use in the design and manufacture of military equipment under the jurisdiction of the Department of Defense.

(b) Guides for the choice and application of switches and associated hardware for use in military equipment.

Complete detailed requirements for switches and associated hardware listed in this standard are covered in the applicable specification sheets or MS sheets (see 4.2). (The complete part number or type designation, and the number of the applicable specification sheet or MS sheet should be referenced in procurement documents for the switches or associated hardware.) When it has been determined that circuit requirements cannot be met by using the switch styles or characteristics listed in this standard, the design engineer with the approval of the cognizant military activity should select from the applicable switch specification, styles or characteristics not listed herein.

1.2 Purpose of standard. The purpose of this standard is to:

(a) Provide the equipment designer with a list of switches and their associated hardware that are considered to be the most flexible for military applications.

(b) Control and minimize the variety of switches used in military equipment in order to facilitate logistic support of the equipment in the field.

(c) Establish criteria pertinent to the choice, application, and use of switches and their associated hardware in military equipment.

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-3786	-	Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification For.
MIL-S-3950	-	Switches, Toggle, Environmentally Sealed, General Specification For.
MIL-B-5423	-	Boots, Dust and Water Seal (For Toggle and Pushbutton Switches, Circuit Breakers, and Rotary-Actuated Parts), General Specification For.
MIL-S-6807	-	Switches, Rotary, Selector Power, General Specification For.
MIL-G-7703	-	Guard, Switch, General Specification For.
MIL-S-8805	-	Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification For.
MIL-S-8834	-	Switches, Toggle, Positive Break, General Specification For.
MIL-S-9395	-	Switches, Pressure, (Absolute, Gage, and Differential), General Specification For.
MIL-S-15291	-	Switches, Rotary, Snap Action.
MIL-S-22710	-	Switches, Rotary (Printed Circuit), (Thumbwheel, Inline, and Pushbutton), General Specification For.
MIL-S-22885	-	Switch, Push Button, Illuminated, General Specification For.
MIL-S-24236	-	Switches, Thermostatic, (Metallic and Bimetallic) General Specification For.
MIL-S-24317	-	Switches, Multistation, Pushbutton, (Illuminated and Non-Illuminated), General Specification For.
MIL-S-28788	-	Switches, Air and Liquid Flow, Sensing General Specification For.
MIL-S-28827	-	Switches, Thermostatic, (Volatile Liquid), Hermetically Sealed General Specification For.
MIL-S-55433	-	Switches, Reed, General Specification For.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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(k) MIL-S-24236 - Switches, Thermostatic (Metallic and Bimetallic). Used primarily in ac and dc applications where temperature protection or accurate temperature control of an enclosure is required.
(l) MIL-S-24317 - Switches, Multistation, Pushbutton (Illuminated and Non-Illuminated). Used as panel displays and switching devices in ac and dc applications.
(m) MIL-S-28788 - Switches, Air and Liquid Flow, Sensing. Used primarily in equipment to sense velocity and flow of air or liquid.
(n) MIL-S-28827 - Switches, Thermostatic (Volatile Liquid), Hermetically Sealed. Used primarily in ac and dc applications that require rapid temperature response.
(o) MIL-S-55433 - Switches, Reed. Used in circuits as a separate element where magnetically-actuated, hermetically sealed switch contacts are required. These reed switches are capable of millions of operations and operations with low-level (dry circuit) loads.

6.3 Switch selection. The designer, in selecting a switch type for a particular function to be performed, must weigh numerous factors before coming to a final decision. Selection normally starts with the most important characteristic for the application, then selecting and compromising other characteristics.

6.3.1 Selection factors. Some of the most important of these factors are noted below:

(a) Design and construction:
(1) Open or enclosed construction.
(2) Sealed (hermetically, environmentally (resilient), dust-tight, water-tight, explosion).
(3) Mounting (bushing, multihole, mounting bracket, mounting plate).
(4) Illuminated or non-illuminated.

(b) Operating characteristics:
(1) Actuation - pushbutton, toggle, rotary, sensing.
(2) Switching action - momentary action, maintained action, alternate action, snap action.

(c) Contacts:
(1) Type of contacts and contact arrangement.
(2) Contact ratings - resistive, inductive, lamp, motor, low-level (dry circuit), and frequency.
(3) Contact bounce time.
(4) Contact resistance.
(5) Make-before-break, break-before-make; shorting or non-shorting contacts.

(d) Environmental considerations:
(1) Temperature range.
(2) Moisture.
(3) Altitude.
(4) Shock and vibration.
(5) Acceleration.
(6) Sand and dust.
(7) Explosion.

(e) Insulation requirements:
(1) Insulation resistance.
(2) Breakdown voltage.

(f) Switching speed:
(1) Actuation speed.

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of 20 June 1977

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(g) Life:

(1) Mechanical.
(2) Electrical at elevated temperature.

(h) Terminals:

(1) Solder.
(2) Screw.
(3) Wire.
(4) Plug-in termination.
(5) Integrated wire termination.
(6) Printed-circuit board terminations.

6.4 Switch parameters.

6.4.1 Design and construction.

(a) Design. Design of the switch, related to available space, mounting dimensions, handling of the actuating element, type of connections to the switch, and aspects relating to ruggedness and safety, is an important selection factor.

(b) Construction. Many types of enclosures are available to protect the switch from external conditions, particularly high humidity and dirt. Switches may be classified according to the degree of protection offered by the enclosure. Such classifications include the following: Open, sealed, enclosed, environmentally (resilient), and hermetically sealed. With the open construction switch, no effort is made to protect the switch or its parts from atmospheric conditions. The enclosed switch is one in which the contacts are enclosed in a closed case made of plastic or metal and plastic. The environmentally (resilient) sealed switch is one in a completely sealed case where any portion of the seal is a resilient material such as a gasket or a seal in the bushing of a panel-mounted switch. The hermetically sealed switch is made airtight by a sealing process which involves fusing or soldering and does not use a gasket. The hermetically sealed enclosure offers the greatest protection because it insulates against such elements as moisture, harmful gases, and dirt. It also eliminates the increased arcing by low atmospheric pressures at high altitudes.

6.4.2 Contacts. The contacts which carry the electrical load can be classified by function, current-carrying capacity, and application. The contact arrangements vary in complexity from a simple make or break, through make-before-break, break-before-make, make-make, break-break, etc, and from single throw to multiple throw, and single pole to multipole and various combinations of these features.

6.4.2.1 Contact ratings. Contacts are usually given multiple ratings dependent on the type of load being switched. These ratings consist of resistive, low-level (dry circuit), lamp, motor, or inductive loads. Most switches are given the resistive load rating and in most instances at least one additional of the above ratings.

6.4.2.2 Contact operate and bounce times. In many applications, critical operate and bounce times of the contacts are important. Operate time in a double-throw switch is defined as the time it takes the moving contact to separate from the normally closed contact, travel to the normally open contact and make the circuit (this does not include bounce time). Bounce time is the interval between first make of the contact until the uncontrolled making and breaking of the contact ceases. In many electronic circuits, a millisecond is a long time and operate and bounce times become critical parameters.

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SECTION 100

SWITCHES

Section

101 Switches, Reed

102 Switches, Push Button, Illuminated

103 Switches, Interlock

104 Switches, Multistation

105 Switches, Pressure

106 Switches, Push Button

107 Switches, Rotary

108 Switches, Rotary, Electro-Mechanical Actuated (Solenoid Type)

109 Switches, Rotary, Printed Circuit (Thumbwheel and Pushbutton)

110 Switches, Sensitive

111 Switches, Thermostatic

112 Switches, Toggle

113 Switch Assemblies, Modular

114 Switches, Air and Liquid Flow

Supersedes page 100.1
of 23 March 1976

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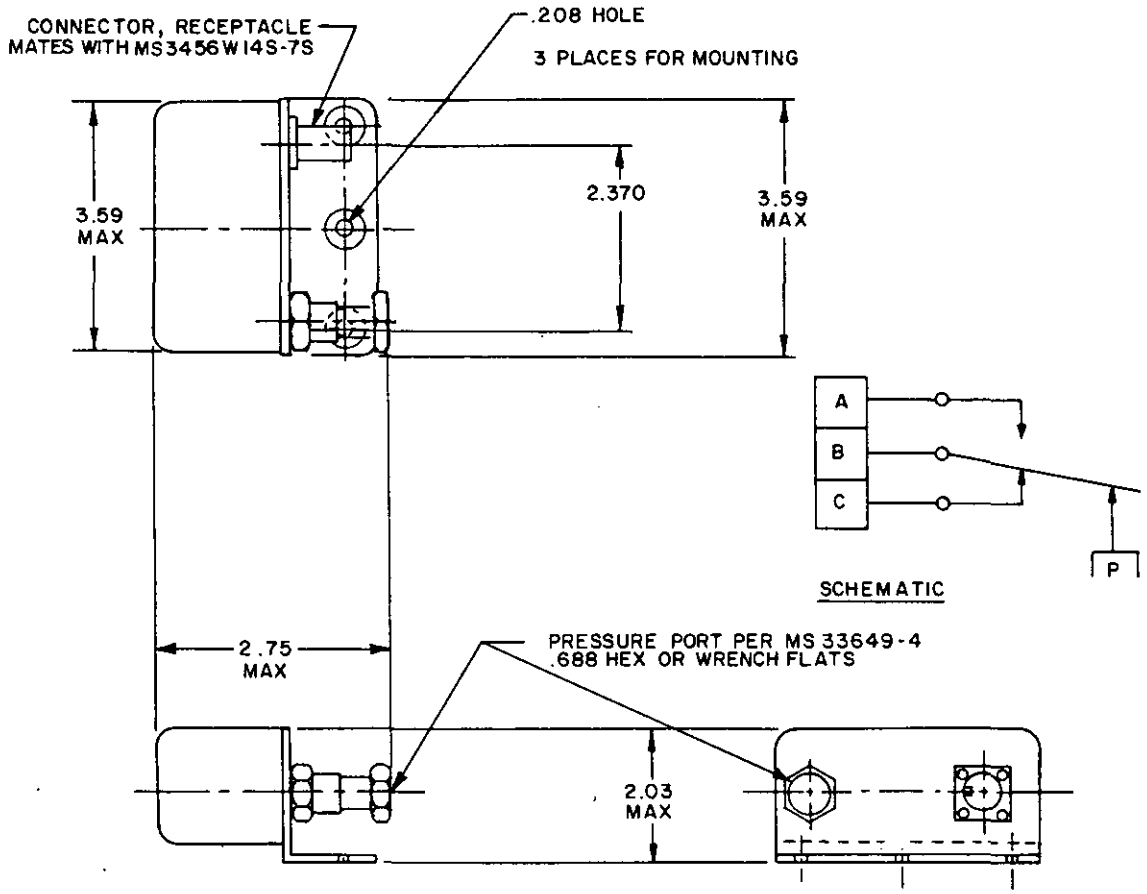
SECTION 105A

SWITCHES, PRESSURE
(Applicable specification: MIL-S-9395)

FIGURE AND SPEC SHEET NO.	SWITCH CHARACTERISTICS					PAGE
	Type	Electrical rating at 28 Vdc (amperes)	System pressure	Proof pressure	Pressure medium	
105-1 - MIL-S-9395/1	Gage	3 resistive	3100 psig	4500 psig	Hydraulic fluid	105.2
105-2 - MIL-S-9395/2	Gage	4 resistive	3000 lb_f/in^2	4500 lb_f/in^2	Hydraulic fluid	105.3
105-3 - MIL-S-9395/5	Gage	5 resistive	3000 psig	4500 psig	Hydraulic fluid	105.4
105-4 - MIL-S-9395/6	Absolute	3 resistive	100 psia	150 psia	Fuel	105.5
105-5 - MIL-S-9395/10	Gage	10 resistive	500 psig	800 psig	Oil	105.6
105-6 - MIL-S-9395/26	Absolute	3 resistive	60 lb_f/in^2	120 lb_f/in^2	Dry air, nitrogen, fuel, lube oil, or hydraulic fluid	105.8
105-7 - MIL-S-9395/27	Gage	3 resistive	60 lb_f/in^2	120 lb_f/in^2	Dry air, nitrogen, fuel, lube oil, or hydraulic fluid	105.10
105-8 - MIL-S-9395/29	Differential	4 resistive	125 psig high over low and simultaneously	250 psig high over low and simultaneously	Dry air, nitrogen, fuel, lube oil, or hydraulic fluid	105.12
105-9 - MIL-S-9395/30	Gage	5 inductive	3000 lb_f/in^2	4500 lb_f/in^2	Hydraulic fluid	105.14
105-10 - MIL-S-9395/31	Gage	5 resistive	3000 psig	4500 psig	Hydraulic fluid, fuel, or lube oil	105.18
105-11 - MIL-S-9395/32	Differential	4 resistive	125 psig high over low and simultaneously	250 psig high over low and simultaneously	Dry air, nitrogen, fuel, oil, or hydraulic fluid	105.22
105-12 - MIL-S-9395/35	Absolute	3 resistive	60 lb_f/in^2	120 lb_f/in^2	Dry air, nitrogen, fuel, lube oil, or hydraulic fluid	105.25
105-13 - MIL-S-9395/38	Gage	5A & low level	125 lb_f/in^2	250 lb_f/in^2	Dry air, nitrogen, fuel, lube oil, hydraulic fluid, oxygen, Coolanol 25R or equal	105.27
105-14 - MIL-S-9395/39	Gage	Low level to 5A	180 lb_f/in^2	270 lb_f/in^2	Dry air, nitrogen, fuel, lube oil, hydraulic fluid, oxygen, Coolanol 25R or equal	105.29
105-15 - MIL-S-9395/40	Gage	Low level to 5A	3000 lb_f/in^2	4500 lb_f/in^2	Hydraulic fluid	105.31
105-16 - MIL-S-9395/41	Gage	Low level to 1A	180 lb_f/in^2	270 lb_f/in^2	Dry air, nitrogen, oxygen, Coolanol 25R or equal	105.33

Supersedes Section 105 (pages 105.1 thru 105.24) of 23 March 1976

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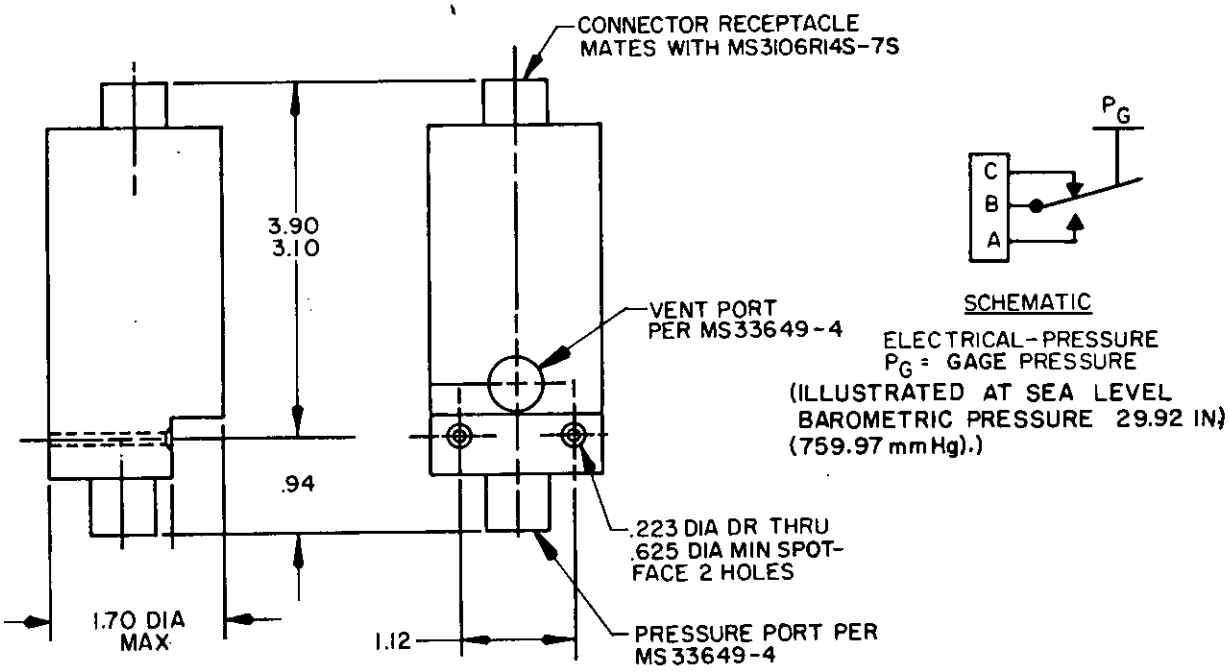


Part number	Switch setting on increasing pressure (psig)	Switch setting on decreasing pressure (psig)	Burst pressure (psig)
M9395/1-01	1500 ±50	1300 ±100	7500
-02	Before 700	350 ±50	7500
-03	2850 ±100	2550 $^{+100}_{-50}$	7500

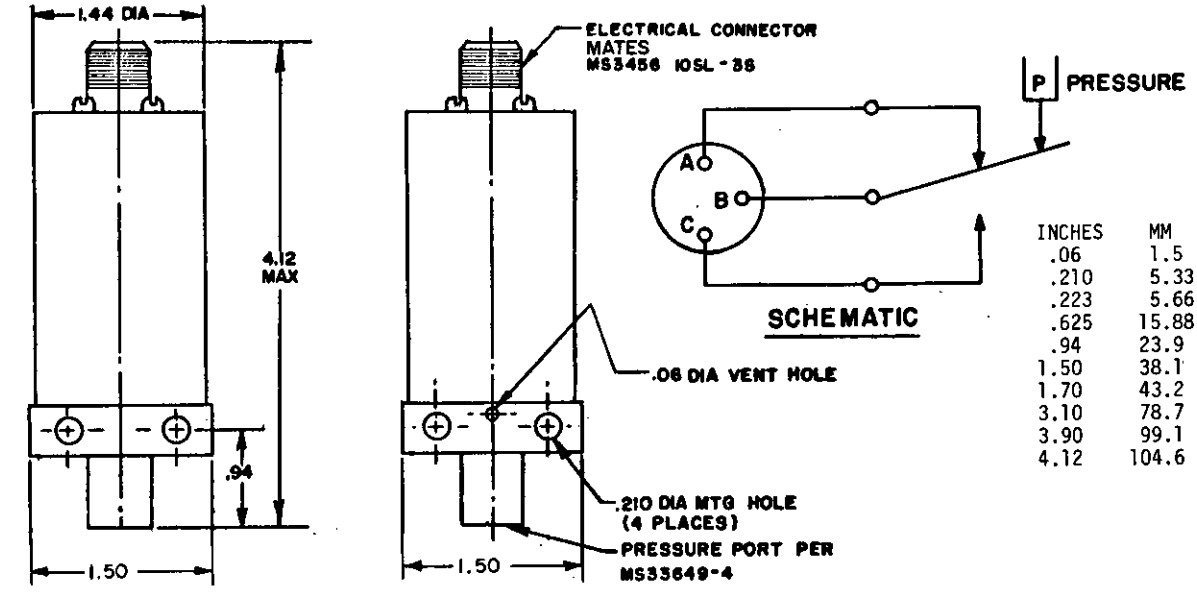
INCHES	MM
.208	5.28
.688	17.48
2.03	51.6
2.370	60.20
2.75	69.8
3.59	91.2

FIGURE 105-1. Gage switches (MIL-S-9395/1).

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CONFIGURATION 1



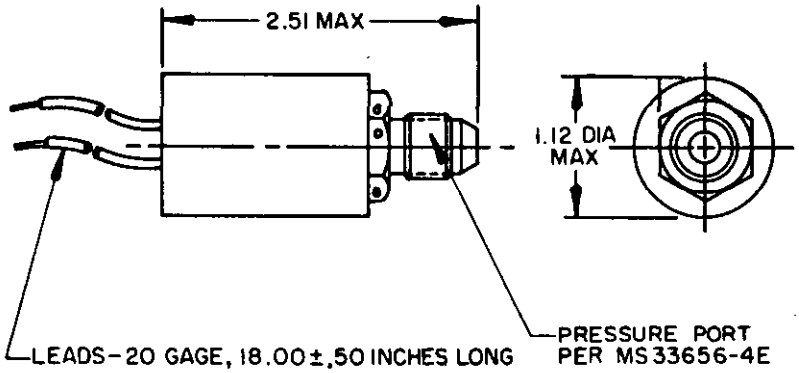
INCHES	MM
.06	1.5
.210	5.33
.223	5.66
.625	15.88
.94	23.9
1.50	38.1
1.70	43.2
3.10	78.7
3.90	99.1
4.12	104.6

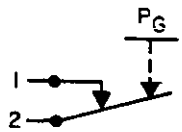
CONFIGURATION 2

CONFIGURATION 1			CONFIGURATION 2		
Part no.	Pressure switch settings (lb_f/in^2)		Part no.	Pressure switch settings (lb_f/in^2)	
	Increasing	Decreasing		Increasing	Decreasing (±100)
M9395/2-01 M9395/2-02	1200 +250 2900 ±150	By 700 By 2400	M9395/2-03 M9395/2-04	By 2000 By 2875	At 1700 At 2600

FIGURE 105-2. Gage switches, burst pressure 7500 lb_{f/in^2} (MIL-S-9395/2).

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SCHEMATIC

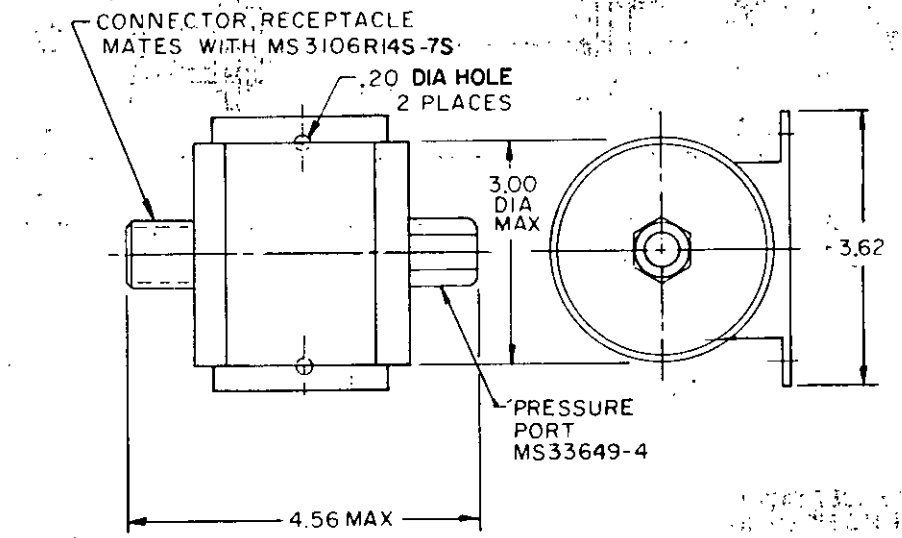
(ILLUSTRATED AT SEA LEVEL BAROMETRIC PRESSURE 29.92 IN. (759.97 mm Hg).)

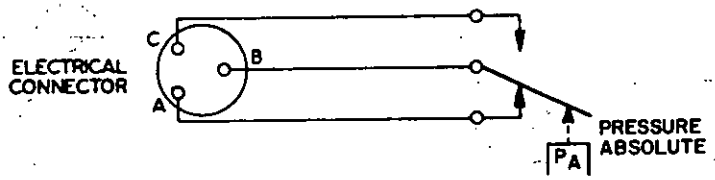
INCHES	MM
.50	12.70
1.12	28.45
2.51	63.75
18.00	457.20

Part number	Switch setting on increasing pressure (psig)	Switch setting on decreasing pressure (psig)	Burst pressure (psig)
M9395/5-01	900 ±50	675 min	7500

FIGURE 105-3. Gage switch (MIL-S-9395/5).

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INCHES	MM
.20	5.1
3.00	76.2
3.62	91.9
4.56	115.8

WIRING SCHEMATIC

Part number	Switch setting on increasing pressure (psia)	Switch setting on decreasing pressure (psia)	Burst pressure (psia)
M9395/6-01	72^{+0}_{-1}	70 ±1	260
-02	52^{+0}_{-1}	50 ±1	260

FIGURE 105-4. Absolute switches (MIL-S-9395/6).

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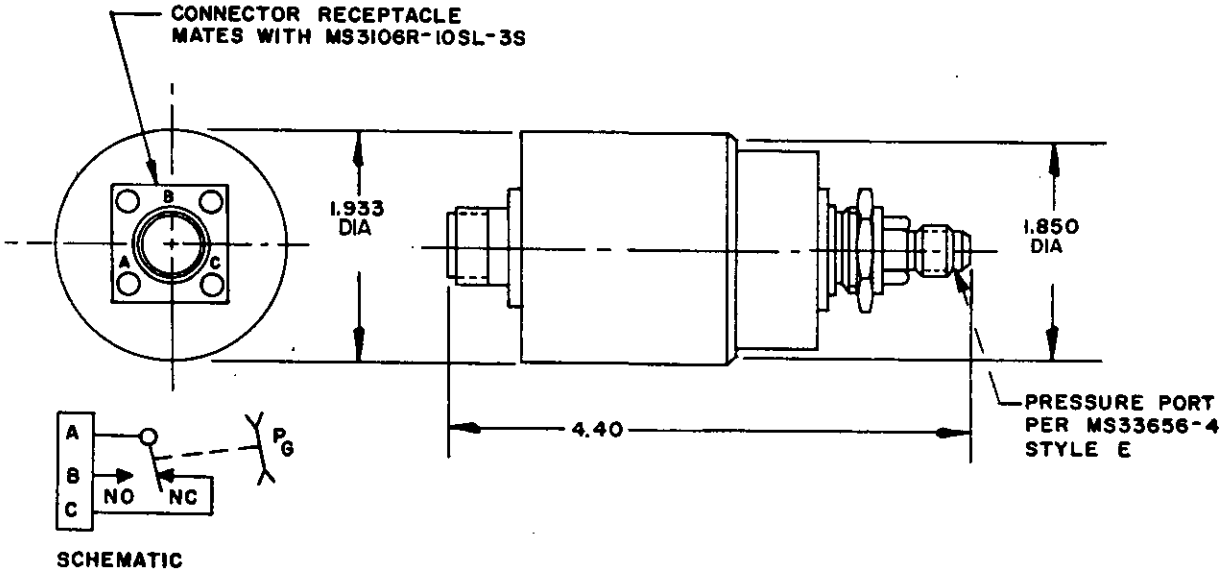


FIGURE 105A (part number M9395/10-01).

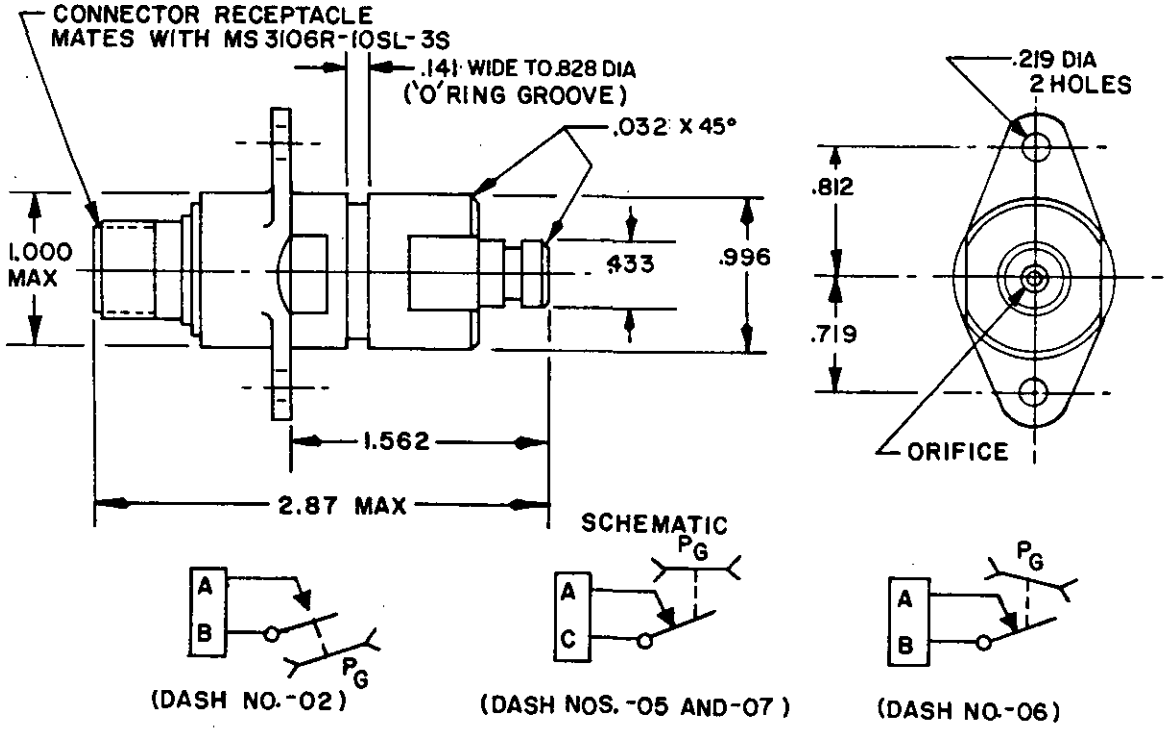


FIGURE 105B (part numbers M9395/10-02, -05, -06, and -07).

FIGURE 105-5. Gage switches (MIL-S-9395/10).

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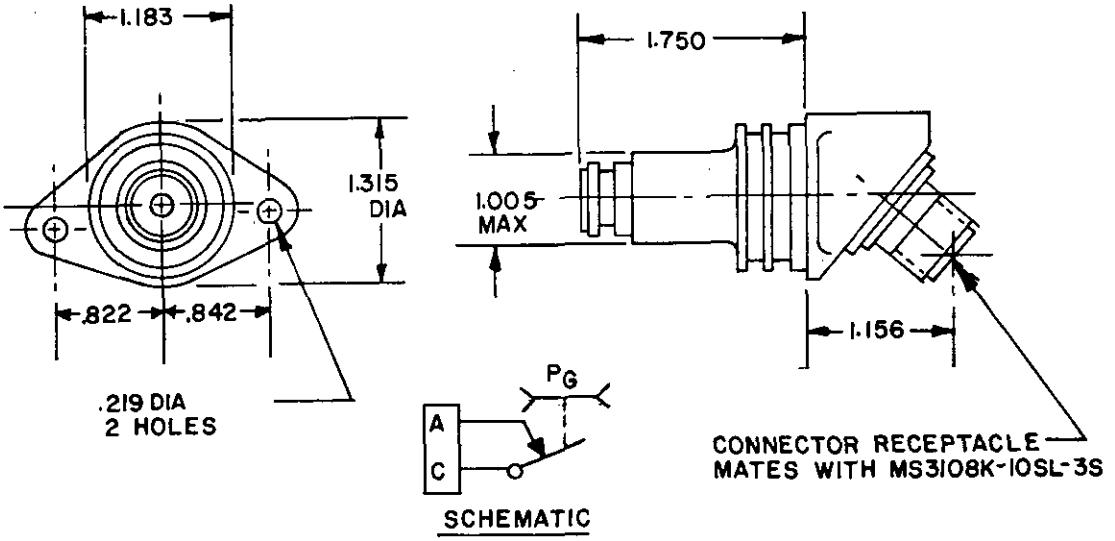


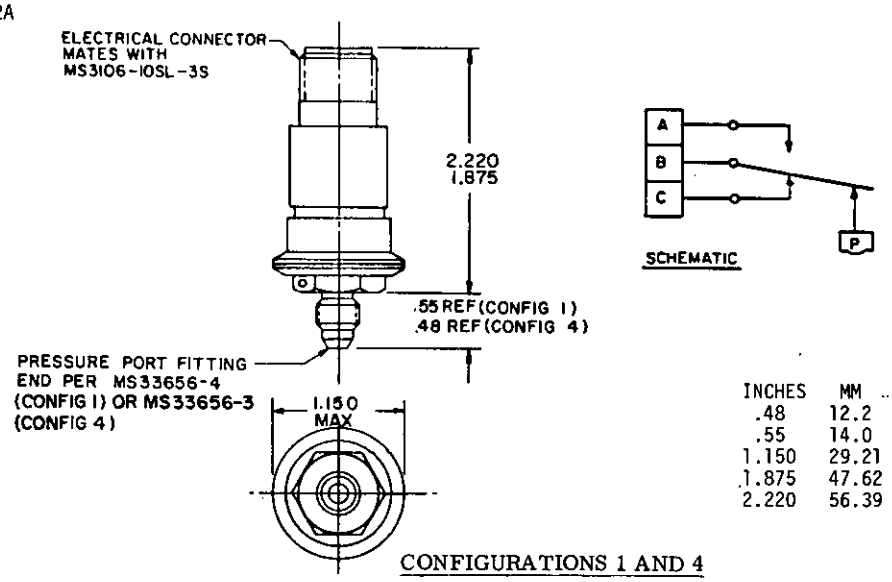
FIGURE 105C (part number M9395/10-03).

INCHES	MM	INCHES	MM	INCHES	MM
.032	.81	.828	21.03	1.315	33.40
.141	3.58	.842	21.39	1.562	39.67
.219	5.56	.996	25.30	1.750	44.45
.433	11.00	1.000	25.40	1.850	46.99
.719	18.26	1.005	25.53	1.933	49.10
.812	20.62	1.156	29.36	2.87	72.9
.822	20.88	1.183	30.05	4.40	111.8

Part number	Switch setting on increasing pressure (psig)	Switch setting on decreasing pressure (psig)	Burst pressure (psig)
M9395/10-01	85 ±20	60 ±10	1200
-02	150 max	50 min	1200
-03	150 max	50 min	1200
-05	150 max	50 min	1200
-06	150 max	50 min	1200
-07	150 max	50 min	1200

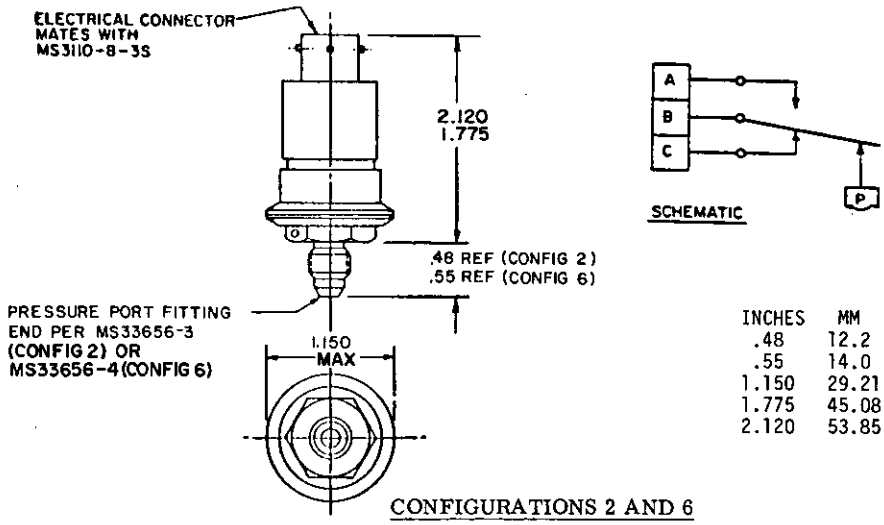
FIGURE 105-5. Gage switches (MIL-S-9395/10) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



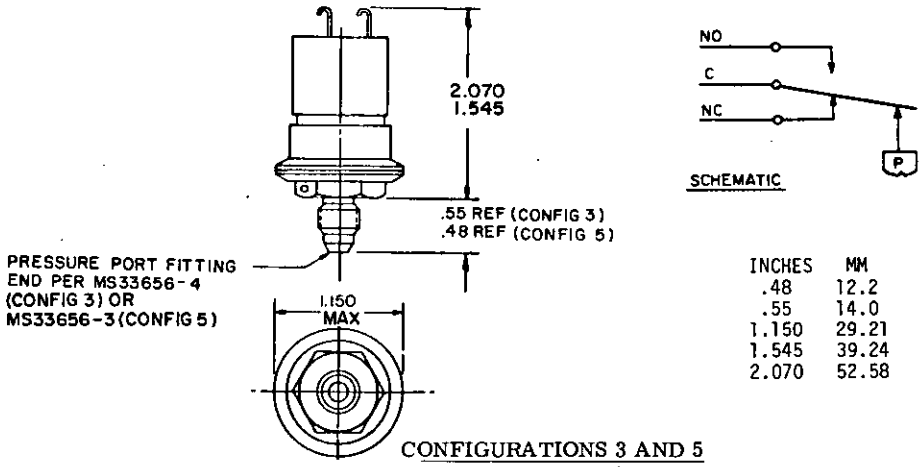
INCHES	MM
.48	12.2
.55	14.0
1.150	29.21
1.875	47.62
2.220	56.39

CONFIGURATIONS 1 AND 4



INCHES	MM
.48	12.2
.55	14.0
1.150	29.21
1.775	45.08
2.120	53.85

CONFIGURATIONS 2 AND 6



INCHES	MM
.48	12.2
.55	14.0
1.150	29.21
1.545	39.24
2.070	52.58

CONFIGURATIONS 3 AND 5

NOTE: Schematics shown are for switches with pressure ports exposed to room ambient.

FIGURE 105-6. Absolute switches (MIL-S-9395/26).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

PRESSURE SETTING (WITH TOLERANCE) AND MAXIMUM DIFFERENTIAL VALUES FOR EACH SETTING:

Pressure setting (lb_f/in^2)		Max differential (lb_f/in^2)	Pressure setting (lb_f/in^2)		Max differential (lb_f/in^2)
Value	Tolerance (±)		Value	Tolerance (±)	
1.0 - 10.5	.75	2.5	44.5	1.5	4.6
11.0 - 20.0	.75	2.5	45.0 - 45.5	2.0	4.7
20.5 - 29.5	1.0	3.0	46.0 - 46.5	↓	4.8
30.0 - 30.5	1.0	3.1	47.0 - 47.5	↓	4.9
31.0	1.5	3.2	48.0 - 48.5	↓	5.0
31.5 - 32.0	↓	3.3	49.0 - 49.5	↓	5.1
32.5 - 33.0	↓	3.4	50.0 - 50.5	↓	5.2
33.5 - 34.0	↓	3.5	51.0 - 51.5	↓	5.3
34.5 - 35.0	↓	3.6	52.0 - 52.5	↓	5.4
35.5 - 36.0	↓	3.7	53.0 - 53.5	↓	5.5
36.5 - 37.0	↓	3.8	54.0 - 54.5	↓	5.6
37.5 - 38.0	↓	3.9	55.0 - 55.5	↓	5.7
38.5 - 39.0	↓	4.0	56.0 - 56.5	↓	5.8
39.5 - 40.0	↓	4.1	57.0 - 57.5	↓	5.9
40.5 - 41.0	↓	4.2	58.0 - 58.5	↓	6.0
41.5 - 42.0	↓	4.3	59.0 - 59.5	↓	6.1
42.5 - 43.0	↓	4.4	60.0	2.0	6.2
43.5 - 44.0	1.5	4.5			

PART NUMBER: Consists of "M9395/26" followed by a dash and a three-letter code. The three-letter code in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 18.0 ±.75 lb_f/in^2 with a maximum differential of 2.5 lb_f/in^2

EXAMPLE:

M9395/26- A G E

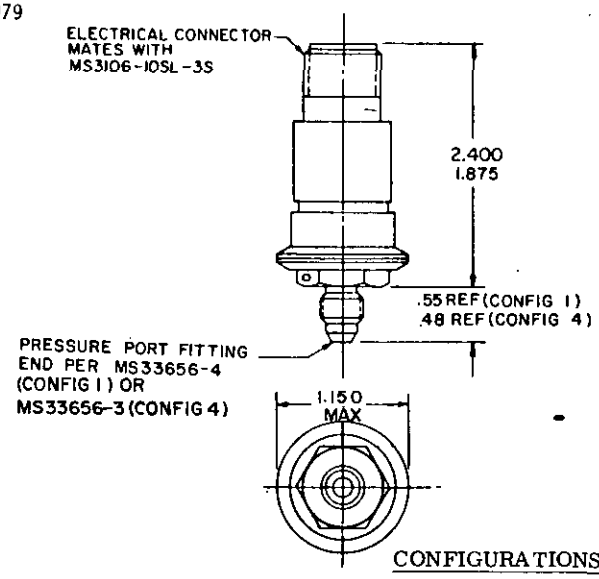
CONFIGURATION AND CONTACT ACTION						
	Configuration					
	1	2	3	4	5	6
Actuates on increasing pressure	A	C	E	G	J	L
Actuates on decreasing pressure	B	D	F	H	K	M

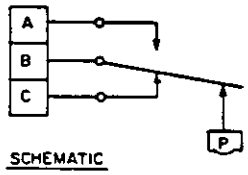
PRESSURE SETTING TO WITHIN 1 lb_f/in^2					
Pressure	Code	Pressure	Code	Pressure	Code
2.0	A	22.0	J	42.0	S
4.5	B	24.5	K	44.5	T
7.0	C	27.0	L	47.0	U
9.5	D	29.5	M	49.5	V
12.0	E	32.0	N	52.0	W
14.5	F	34.5	P	54.5	X
17.0	G	37.0	Q	57.0	Y
19.5	H	39.5	R	59.5	Z

PRESSURE SETTING TO WITHIN .5 lb_f/in^2			
Unit	Code	Unit	Code
-1.0	A	0.5	D
-0.5	B	1.0	E
0	C		

FIGURE 105-6. Absolute switches (MIL-S-9395/26) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

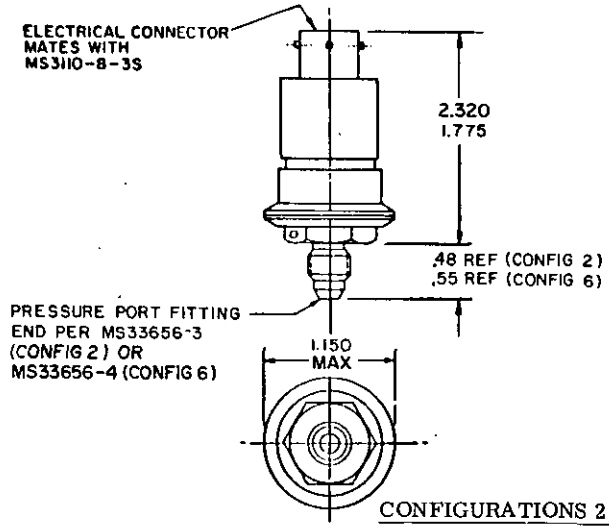


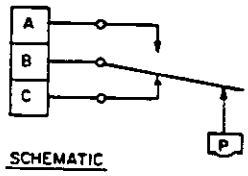


SCHEMATIC

INCHES	MM
.48	12.2
.55	14.0
1.150	29.21
1.875	47.62
2.400	60.96

CONFIGURATIONS 1 AND 4

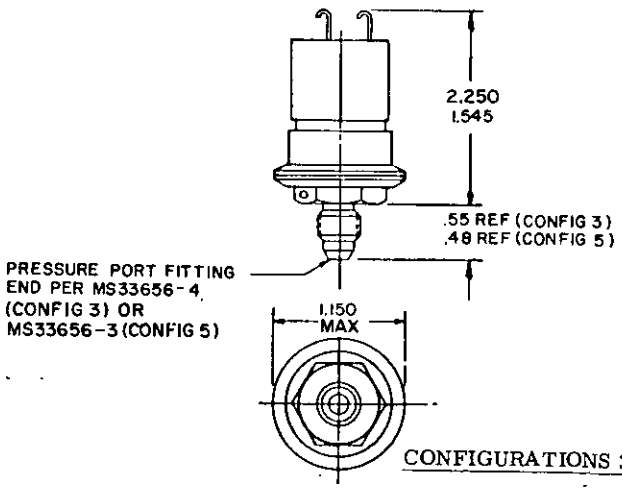


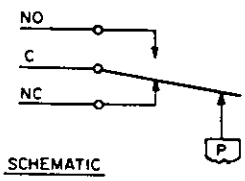


SCHEMATIC

INCHES	MM
.48	12.2
.55	14.0
1.150	29.21
1.775	45.08
2.320	58.93

CONFIGURATIONS 2 AND 6





SCHEMATIC

INCHES	MM
.48	12.2
.55	14.0
1.150	29.21
1.545	39.24
2.250	57.15

CONFIGURATIONS 3 AND 5

NOTE: Schematics shown are for switches with pressure ports exposed to room ambient.

FIGURE 105-7. Gage switches (MIL-S-9395/27).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

PRESSURE SETTING (WITH TOLERANCE) AND MAXIMUM DIFFERENTIAL VALUES FOR EACH SETTING:

Pressure setting (lb_f/in^2)		Max differential (lb_f/in^2)	Pressure setting (lb_f/in^2)		Max differential (lb_f/in^2)
Value	Tolerance (±)		Value	Tolerance (±)	
1.0 - 10.5	.50	2.0	44.5	1.5	4.6
11.0 - 20.0	.75	2.5	45.0 - 45.5	2.0	4.7
20.5 - 29.5	1.0	3.0	46.0 - 46.5	↓	4.8
30.0 - 30.5	1.0	3.1	47.0 - 47.5	↓	4.9
31.0	1.5	3.2	48.0 - 48.5	↓	5.0
31.5 - 32.0	↓	3.3	49.0 - 49.5	↓	5.1
32.5 - 33.0	↓	3.4	50.0 - 50.5	↓	5.2
33.5 - 34.0	↓	3.5	51.0 - 51.5	↓	5.3
34.5 - 35.0	↓	3.6	52.0 - 52.5	↓	5.4
35.5 - 36.0	↓	3.7	53.0 - 53.5	↓	5.5
36.5 - 37.0	↓	3.8	54.0 - 54.5	↓	5.6
37.5 - 38.0	↓	3.9	55.0 - 55.5	↓	5.7
38.5 - 39.0	↓	4.0	56.0 - 56.5	↓	5.8
39.5 - 40.0	↓	4.1	57.0 - 57.5	↓	5.9
40.5 - 41.0	↓	4.2	58.0 - 58.5	↓	6.0
41.5 - 42.0	↓	4.3	59.0 - 59.5	↓	6.1
42.5 - 43.0	↓	4.4	60.0	2.0	6.2
43.5 - 44.0	1.5	4.5			

PART NUMBER: Consists of "M9395/27" followed by a dash and a three-letter code. The three-letter code in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 18.0 ±.75 lb_f/in^2 with a maximum differential of 2.5 lb_f/in^2.

EXAMPLE:

M9395/27- A G E

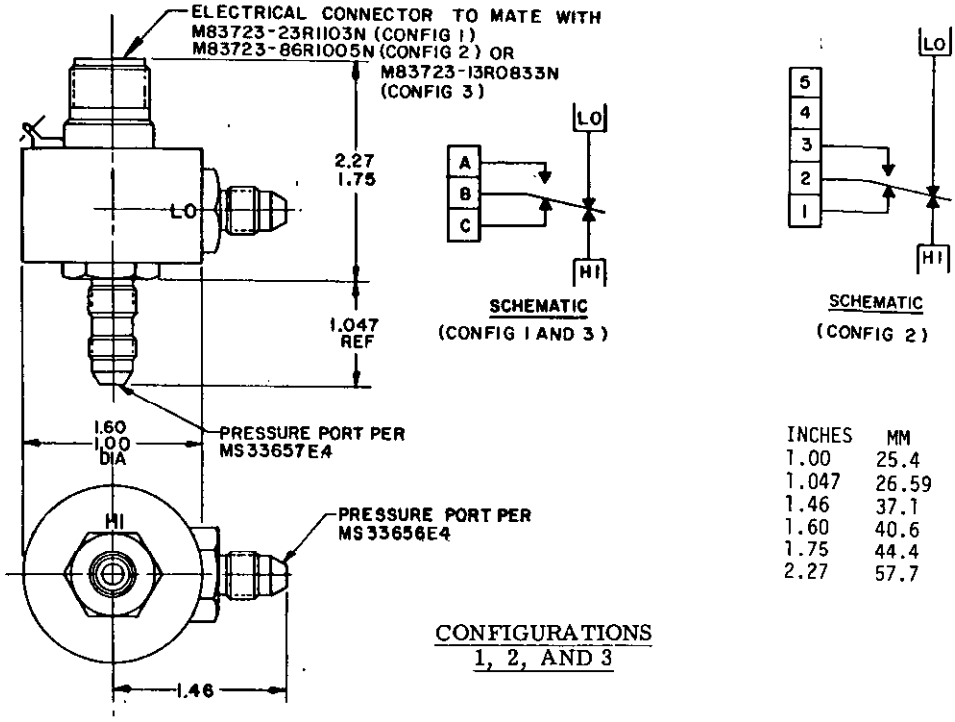
CONFIGURATION AND CONTACT ACTION													
Actuates on increasing pressure							Actuates on decreasing pressure						
Electrical chamber	Configuration						Electrical chamber	Configuration					
	1	2	3	4	5	6		1	2	3	4	5	6
Hermetic - - -	A	E	J	N	S	W	Hermetic - - -	B	F	K	P	T	X
Unsealed - - -	C	G	L	Q	U	Y	Unsealed - - -	D	H	M	R	V	Z

PRESSURE SETTING TO WITHIN 1 lb_f/in^2					
Pressure	Code	Pressure	Code	Pressure	Code
2.0	A	22.0	J	42.0	S
4.5	B	24.5	K	44.5	T
7.0	C	27.0	L	47.0	U
9.5	D	29.5	M	49.5	V
12.0	E	32.0	N	52.0	W
14.5	F	34.5	P	54.5	X
17.0	G	37.0	Q	57.0	Y
19.5	H	39.5	R	59.5	Z

PRESSURE SETTING TO WITHIN .5 lb_f/in^2			
Unit	Code	Unit	Code
-1.0	A	0.5	D
-0.5	B	1.0	E
0	C		

FIGURE 105-7. Gage switches (MIL-S-9395/27) - Continued.

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Check the source to verify that this is the current version before use.



INCHES	MM
1.00	25.4
1.047	26.59
1.46	37.1
1.60	40.6
1.75	44.4
2.27	57.7

CONFIGURATIONS
1, 2, AND 3

PART NUMBER: Consists of "M9395/29" followed by a dash and a five-letter code. The five-letter code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 1-1/2 ±1/2 psid, and deactuates on decreasing pressure at 1/2 ±1/2 psid.

EXAMPLE:

M9395/29- A A D A B

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "A" of table II (coarse value of 0 psid) and Code "D" of table III (fine value of 1-1/2 ±1/2 psid).

LOW-PRESSURE SETTING = Code "A" of table II (coarse value of 0 psid) and Code "B" of table III (fine value of 1/2 ±1/2 psid).

TABLE I. Codes for combinations of configurations and pressure setting modes.

	Configuration 1		Configuration 2		Configuration 3		Pressure setting mode	
	Electrical chamber		Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed	Hermetic	Unsealed		
Code - - -	A	D	G	K	N	R	At (or max) 1/	At (or min) 1/
Code - - -	B	E	H	L	P	S	At (or max) 1/	Differential 2/
Code - - -	C	F	J	M	Q	T	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

FIGURE 105-8. Differential switches (MIL-S-9395/29).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Codes for coarse settings.

Code	Coarse value (in psid)	Code	Coarse value (in psid)	Code	Coarse value (in psid)
A - - -	0	L - - -	30	W - - -	80
B - - -	2-1/2	M - - -	35	X - - -	85
C - - -	5	N - - -	40	Y - - -	90
D - - -	7-1/2	P - - -	45	Z - - -	95
E - - -	10	Q - - -	50	1 - - -	100
F - - -	12-1/2	R - - -	55	2 - - -	105
G - - -	15	S - - -	60	3 - - -	110
H - - -	17-1/2	T - - -	65	4 - - -	115
J - - -	20	U - - -	70	5 - - -	120
K - - -	25	V - - -	75	6 - - -	125

TABLE III. Codes for combinations of fine settings and tolerance values.

For settings below 20 psid							For settings of 20 psid and above						
	Fine value (in psid)					Tolerance		Fine value (in psid)					Tolerance
	0	1/2	1	1-1/2	2			0	1	2	3	4	
Code - - -	A	B	C	D	E	±1/2 psid	Code - - -	A	B	C	D	E	±1 psid 1/
Code - - -	F	G	H	J	K	±1 psid	Code - - -	F	G	H	J	K	±2 psid 2/
Code - - -	L	M	N	P	Q	±2 psid	Code - - -	L	M	N	P	Q	±3 psid 3/
Code - - -	R	S	T	U	V	±3 psid	Code - - -	R	S	T	U	V	±5 psid
Code - - -	W	X	Y	Z	1	±5 psid	Code - - -	W	X	Y	Z	1	±8 psid
Code - - -	2	3	4	5	6	±8 psid	Code - - -	2	3	4	5	6	±15 psid
Code - - -	7	8	9	0	I	Min or Max	Code - - -	7	8	9	0	I	Min or Max

1/ Not applicable for pressure settings above 33 psid.
2/ Not applicable for pressure settings above 66 psid.
3/ Not applicable for pressure settings above 100 psid.

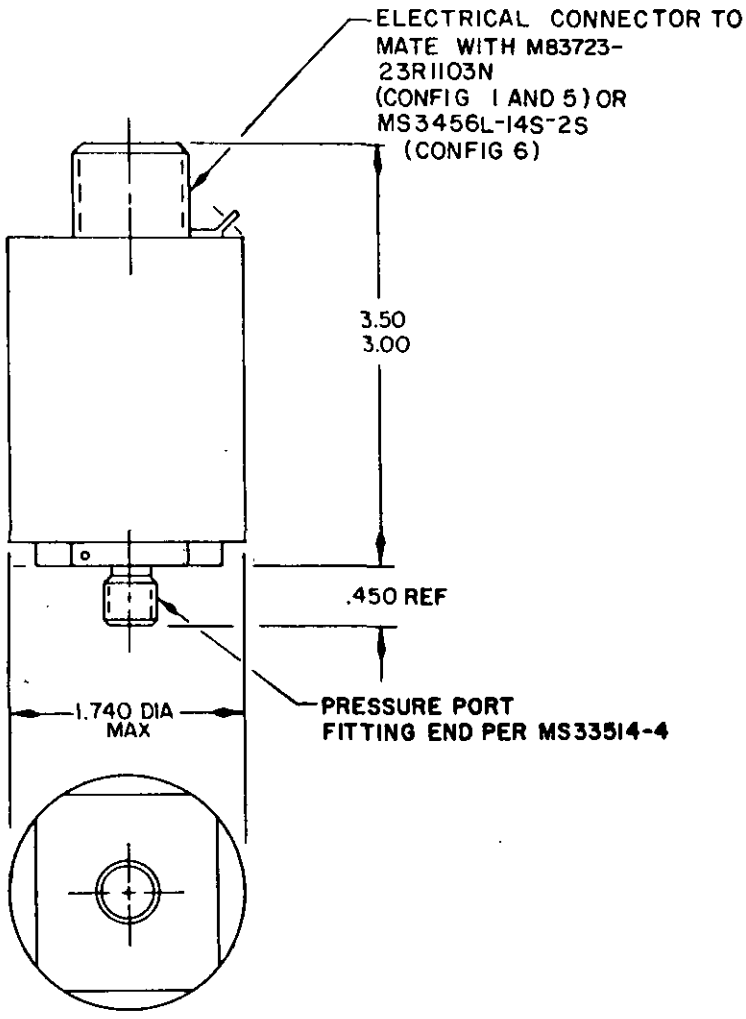
TABLE IV. Codes for differential settings.

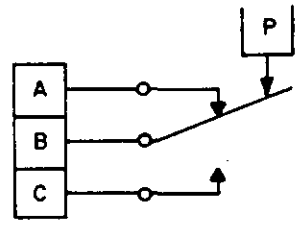
Code	Differential value (in psid)	Code	Differential value (in psid)	Code	Differential value (in psid)
A - - -	0	N - - -	6	1 - - -	20
B - - -	1/2	P - - -	7	2 - - -	22
C - - -	1	Q - - -	8	3 - - -	24
D - - -	1-1/2	R - - -	9	4 - - -	26
E - - -	2	S - - -	10	5 - - -	28
F - - -	2-1/2	T - - -	11	6 - - -	30
G - - -	3	U - - -	12	7 - - -	35
H - - -	3-1/2	V - - -	13	8 - - -	40
J - - -	4	W - - -	14	9 - - -	45
K - - -	4-1/2	X - - -	15	0 - - -	50
L - - -	5	Y - - -	16		
M - - -	5-1/2	Z - - -	18		

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

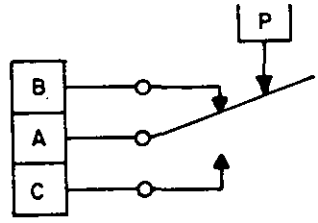
FIGURE 105-8. Differential switches (MIL-S-9395/29) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

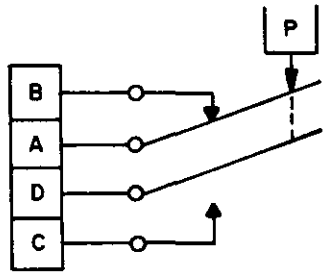




SCHEMATIC
SHOWN AT ZERO PRESSURE
(CONFIGURATION 1)



SCHEMATIC
SHOWN AT ZERO PRESSURE
(CONFIGURATION 5)



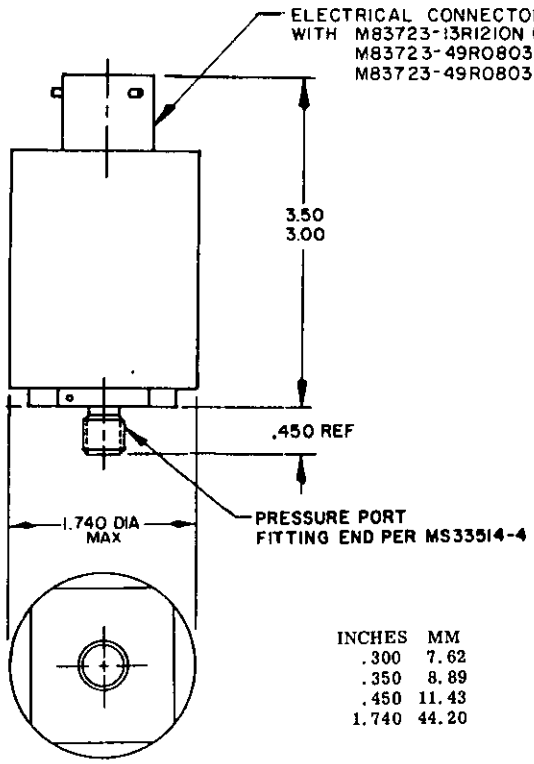
SCHEMATIC
SHOWN AT ZERO PRESSURE
(CONFIGURATION 6)

INCHES	MM
.450	11.43
1.740	44.20
3.00	76.2
3.50	88.9

CONFIGURATIONS 1, 5, AND 6

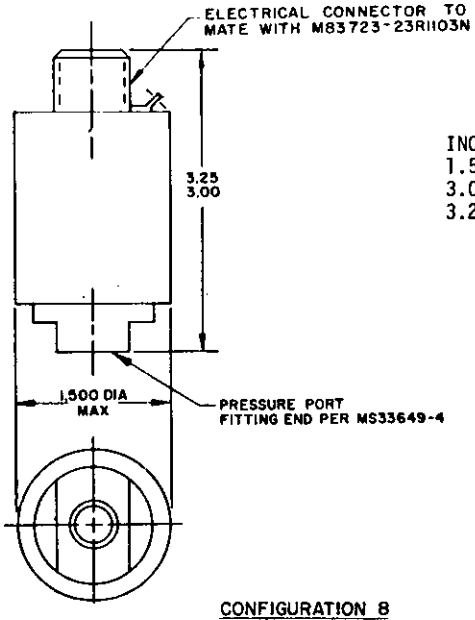
FIGURE 105-9. Gage switches (MIL-S-9395/30).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



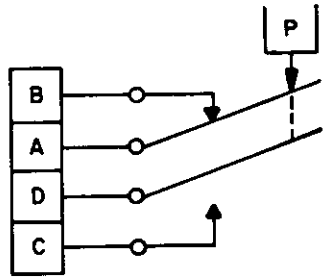
INCHES	MM
.300	7.62
.350	8.89
.450	11.43
1.740	44.20

CONFIGURATIONS 2, 3, 4, AND 7

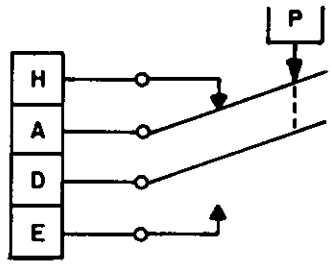


INCHES	MM
1.500	38.10
3.00	76.2
3.25	82.6

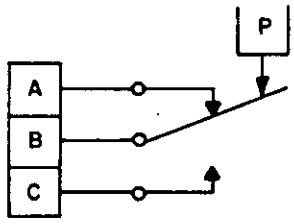
CONFIGURATION 8



SCHEMATIC
SHOWN AT ZERO PRESSURE
(CONFIGURATION 2)



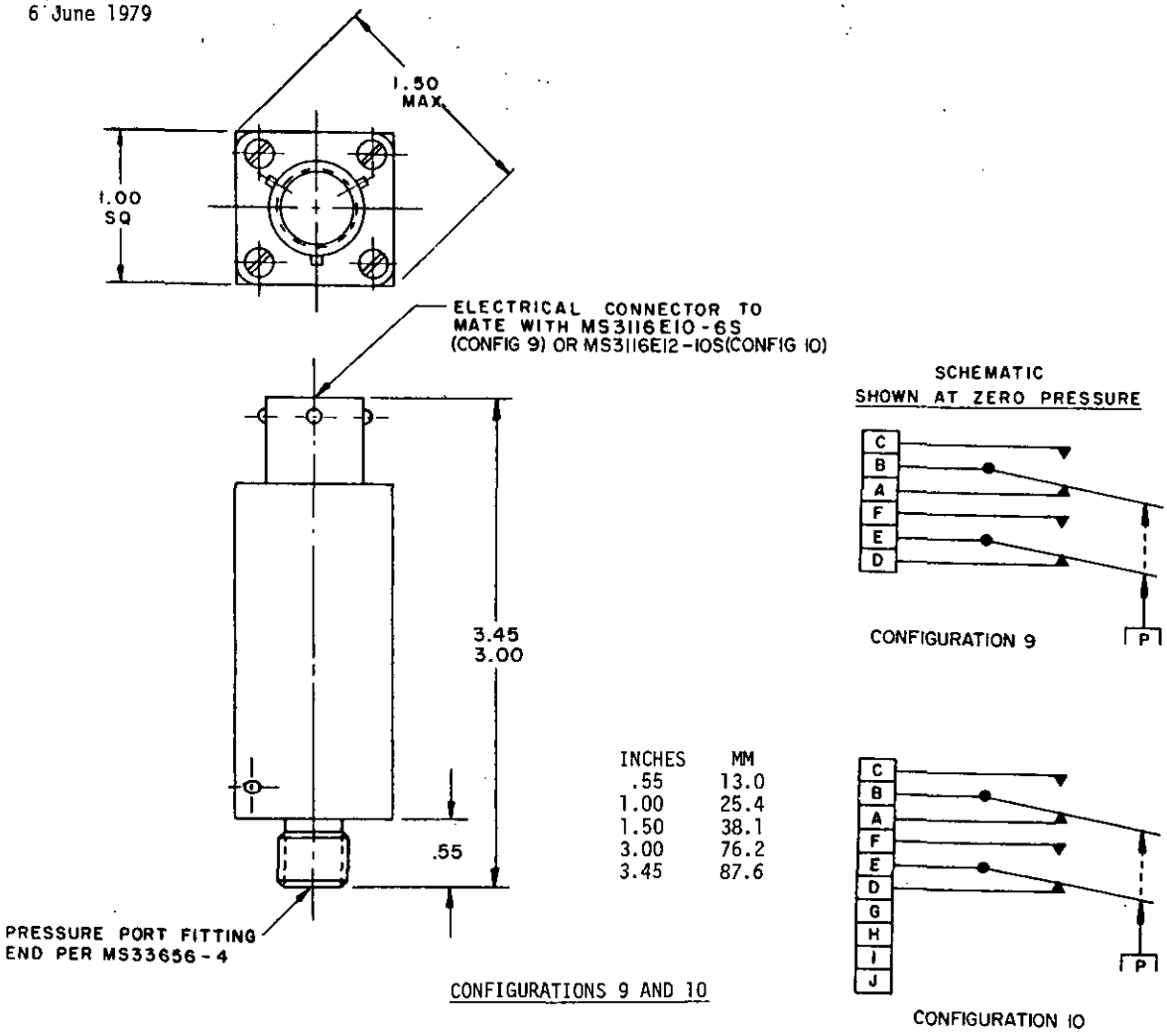
SCHEMATIC
SHOWN AT ZERO PRESSURE
(CONFIGURATION 3)



SCHEMATIC
SHOWN AT ZERO PRESSURE
(CONFIGURATIONS 4,7, AND 8)

FIGURE 105-9. Gage switches (MIL-S-9395/30) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



INCHES	MM
.55	13.0
1.00	25.4
1.50	38.1
3.00	76.2
3.45	87.6

CONFIGURATIONS 9 AND 10

PART NUMBER: Consists of "M9395/30" followed by a dash and a five-letter code. The five-letter code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 300 $\pm$25 lb_f/in^2 and deactuates on decreasing pressure at 150 $\pm$25 lb_f/in^2.

EXAMPLE:

M9395/30- A C A A C

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "C" of table II (300 lb_f/in^2) and Code "A" of table III (0, $\pm$25 lb_f/in^2).

LOW-PRESSURE SETTING = Code "A" of table II (100 lb_f/in^2) and Code "C" of table III (+50, $\pm$25 lb_f/in^2).

FIGURE 105-9. Gage switches (MIL-S-9395/30) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Codes for combinations of configurations and pressure setting modes.

	Configuration										Pressure setting mode	
	1	2	3	4	5	6	7	8	9	10	High setting	Low setting
Code -	A	D	G	K	N	R	U	X	1	4	At (or max) 1/	At (or min) 1/
Code -	B	E	H	L	P	S	V	Y	2	5	At (or max) 1/	Differential 2/
Code -	C	F	J	M	Q	T	W	Z	3	6	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

TABLE II. Codes for pressure settings to within 100 lb_f/in^2.

Code	Pressure (in lb_f/in^2)	Code	Pressure (in lb_f/in^2)	Code	Pressure (in lb_f/in^2)	Code	Pressure (in lb_f/in^2)
A - - -	100	J - - -	900	S - - -	1700	1 - - -	2500
B - - -	200	K - - -	1000	T - - -	1800	2 - - -	2600
C - - -	300	L - - -	1100	U - - -	1900	3 - - -	2700
D - - -	400	M - - -	1200	V - - -	2000	4 - - -	2800
E - - -	500	N - - -	1300	W - - -	2100	5 - - -	2900
F - - -	600	P - - -	1400	X - - -	2200	6 - - -	3000
G - - -	700	Q - - -	1500	Y - - -	2300		
H - - -	800	R - - -	1600	Z - - -	2400		

TABLE III. Codes for pressure settings to within 25 lb_f/in^2 and tolerances.

	Unit				Tolerance (in lb_f/in^2)
	0	+25	+50	+75	
Code - - -	A	B	C	D	±25
Code - - -	E	F	G	H	±50
Code - - -	J	K	L	M	±75
Code - - -	N	P	Q	R	±100
Code - - -	S	T	U	V	±150
Code - - -	W	X	Y	Z	±200
Code - - -	1	2	3	4	±300
Code - - -	5	6	7	8	±400
Code - - -	9	0	I	-	Min or Max

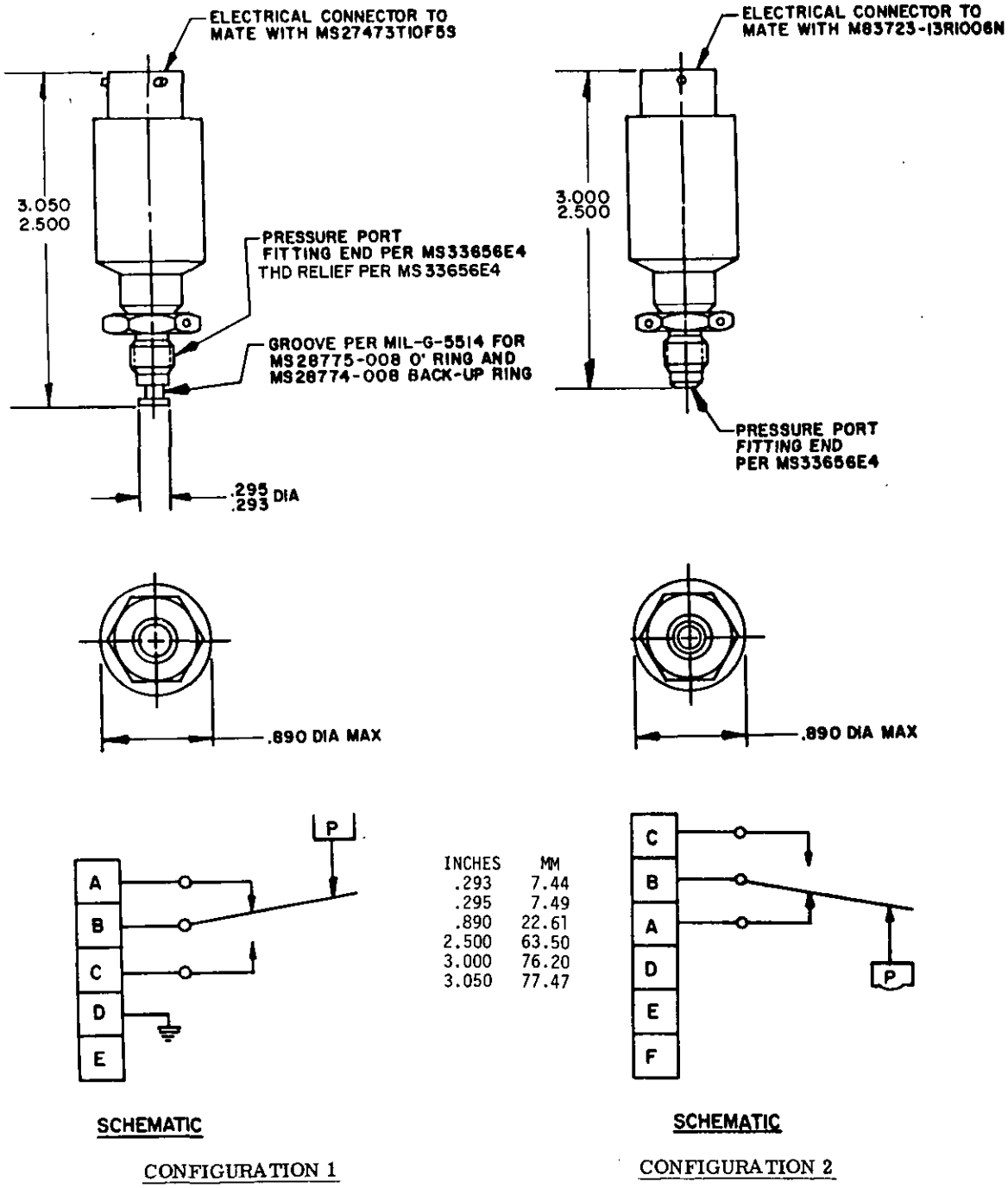
TABLE IV. Codes for differential pressure settings.

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	M - - -	275
B - - -	25	N - - -	300
C - - -	50	P - - -	325
D - - -	75	Q - - -	350
E - - -	100	R - - -	375
F - - -	125	S - - -	400
G - - -	150	T - - -	425
H - - -	175	U - - -	450
J - - -	200	V - - -	475
K - - -	225	W - - -	500
L - - -	250		

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

FIGURE 105-9. Gage switches (MIL-S-9395/30) - Continued.

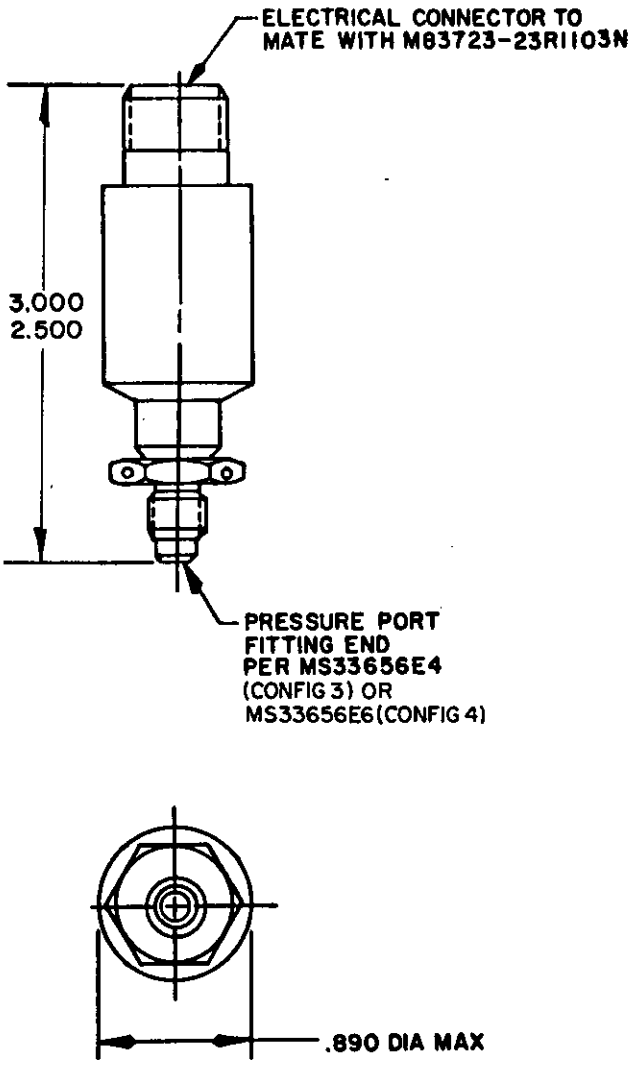
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
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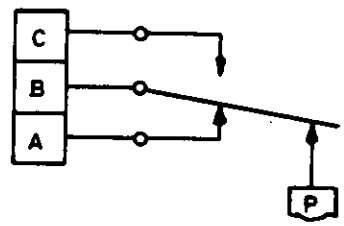


INCHES	MM
.293	7.44
.295	7.49
.890	22.61
2.500	63.50
3.000	76.20
3.050	77.47

FIGURE 105-10. Gage switches (MIL-S-9395/31).

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SCHEMATIC

INCHES	MM
.293	7.44
.295	7.49
.4375	11.112
.5625	14.288
.890	22.61
2.500	63.50
3.000	76.20
3.050	77.47

CONFIGURATIONS 3 AND 4

NOTE: Schematics shown are for switches with pressure ports exposed to room ambient.

FIGURE 105-10. Gage switches (MIL-S-9395/31) - Continued.

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PART NUMBER: Consists of "M9395/31" followed by a dash and a five-letter code. The five-letter code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 300 ±25 psig, and deactuates on decreasing pressure at 150 ±25 psig.

EXAMPLE:

M9395/31- A C A A C

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "C" of table II (300 psig) and Code "A" of table III (0, ±25 psig).

LOW-PRESSURE SETTING = Code "A" of table II (100 psig) and Code "C" of table III (+50, ±25 psig).

TABLE I. Codes for combinations of configurations and pressure setting modes.

	Configuration 1		Configuration 2		Pressure setting mode	
	Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed		
Code - - -	A	D	G	K	At (or max) 1/	At (or min) 1/
Code - - -	B	E	H	L	At (or max) 1/	Differential 2/
Code - - -	C	F	J	M	Differential 2/	At (or min) 1/

	Configuration 3		Configuration 4		Pressure setting mode	
	Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed		
Code - - -	N	R	U	X	At (or max) 1/	At (or min) 1/
Code - - -	P	S	V	Y	At (or max) 1/	Differential 2/
Code - - -	Q	T	W	Z	Differential 2/	At (or min) 1/

1/ Setting values are designated by characters from tables II and III.
2/ Setting values are designated by characters from table V.

TABLE II. Codes for pressure settings to within 100 psig.

Code	Pressure (in psig)	Code	Pressure (in psig)	Code	Pressure (in psig)	Code	Pressure (in psig)
A - - -	100	J - - -	900	S - - -	1700	1 - - -	2500
B - - -	200	K - - -	1000	T - - -	1800	2 - - -	2600
C - - -	300	L - - -	1100	U - - -	1900	3 - - -	2700
D - - -	400	M - - -	1200	V - - -	2000	4 - - -	2800
E - - -	500	N - - -	1300	W - - -	2100	5 - - -	2900
F - - -	600	P - - -	1400	X - - -	2200	6 - - -	3000
G - - -	700	Q - - -	1500	Y - - -	2300		
H - - -	800	R - - -	1600	Z - - -	2400		

FIGURE 105-10. Gage switches (MIL-S-9395/31) - Continued.

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TABLE III. Codes for pressure settings to within 25 psig and tolerances.

	Unit				Tolerance
	0	+25	+50	+75	
Code - - -	A	B	C	D	±25 psig
Code - - -	E	F	G	H	±50 psig
Code - - -	J	K	L	M	±75 psig
Code - - -	N	P	Q	R	±100 psig
Code - - -	S	T	U	V	±150 psig
Code - - -	W	X	Y	Z	±200 psig
Code - - -	1	2	3	4	±300 psig
Code - - -	5	6	7	8	±400 psig
Code - - -	9	0	I	-	Min or Max

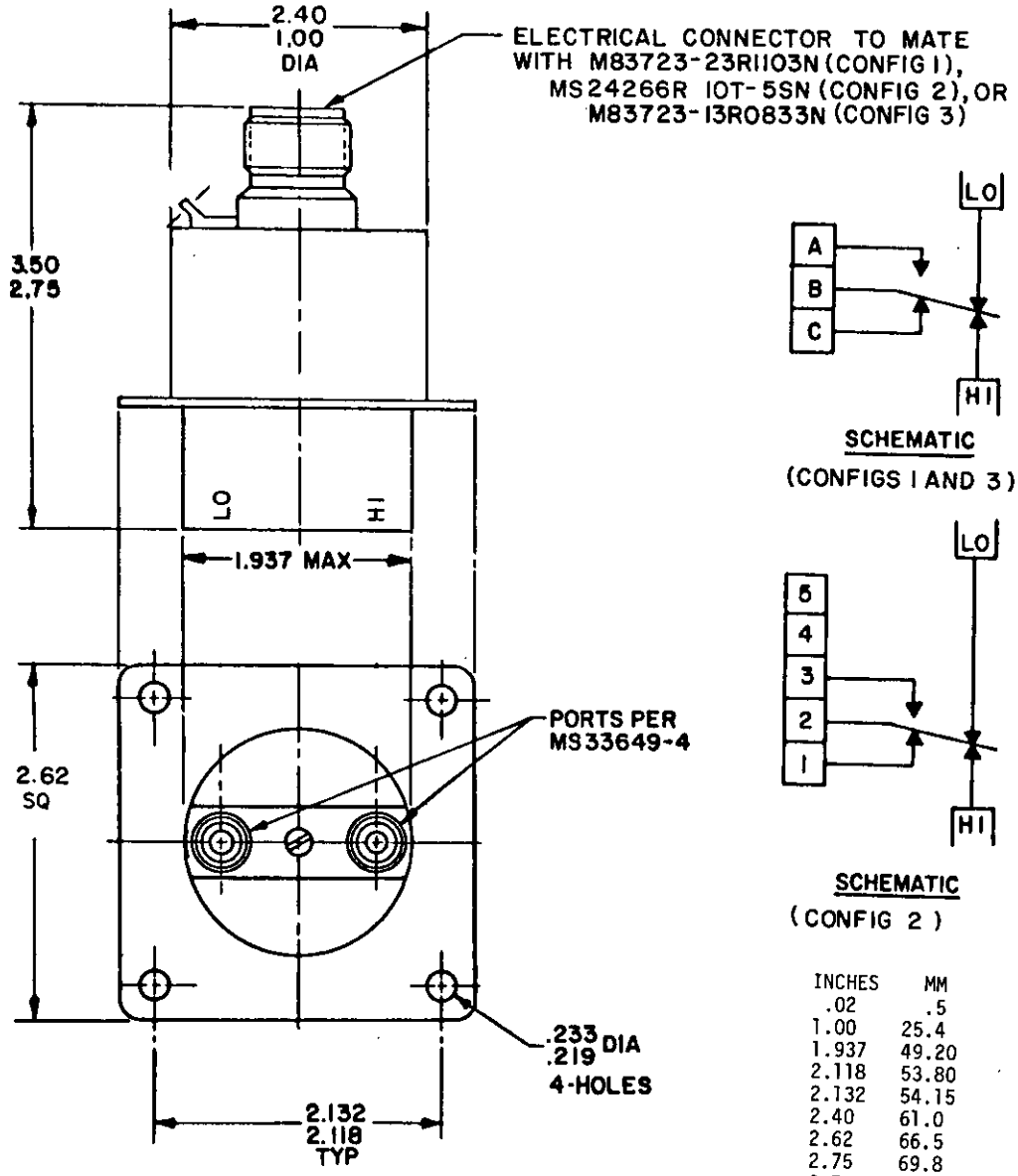
TABLE IV. Codes for differential pressure settings.

Code	Differential value (in psig)	Code	Differential value (in psig)
A - - -	0	M - - -	275
B - - -	25	N - - -	300
C - - -	50	P - - -	325
D - - -	75	Q - - -	350
E - - -	100	R - - -	375
F - - -	125	S - - -	400
G - - -	150	T - - -	425
H - - -	175	U - - -	450
J - - -	200	V - - -	475
K - - -	225	W - - -	500
L - - -	250		

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

FIGURE 105-10. Gage switches (MIL-S-9395/31) - Continued.

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CONFIGURATIONS 1, 2, AND 3

INCHES	MM
.02	.5
1.00	25.4
1.937	49.20
2.118	53.80
2.132	54.15
2.40	61.0
2.62	66.5
2.75	69.8
3.50	88.9

FIGURE 105-11. Differential switches (MIL-S-9395/32).

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PART NUMBER: Consists of "M9395/32" followed by a dash and a five-letter code. The five-letter code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 1-1/2 ±1/2 psid, and deactuates on decreasing pressure at 1/2 ±1/2 psid.

EXAMPLE:

M9395/32- A A D A B

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "A" of table II (coarse value of 0 psid) and Code "D" of table III (fine value of 1-1/2 ±1/2 psid).

LOW-PRESSURE SETTING = Code "A" of table II (coarse value of 0 psid) and Code "B" of table III (fine value of 1/2 ±1/2 psid).

TABLE I. Codes for combinations of configurations and pressure setting modes.

	Configuration 1		Configuration 2		Configuration 3		Pressure setting mode	
	Electrical chamber		Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed	Hermetic	Unsealed		
Code - - -	A	D	G	K	N	R	At (or max) 1/	At (or min) 1/
Code - - -	B	E	H	L	P	S	At (or max) 1/	Differential 2/
Code - - -	C	F	J	M	Q	T	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

TABLE II. Codes for coarse settings.

Code	Coarse value (in psid)	Code	Coarse value (in psid)	Code	Coarse value (in psid)
A - - -	0	L - - -	30	W - - -	80
B - - -	2-1/2	M - - -	35	X - - -	85
C - - -	5	N - - -	40	Y - - -	90
D - - -	7-1/2	P - - -	45	Z - - -	95
E - - -	10	Q - - -	50	1 - - -	100
F - - -	12-1/2	R - - -	55	2 - - -	105
G - - -	15	S - - -	60	3 - - -	110
H - - -	17-1/2	T - - -	65	4 - - -	115
J - - -	20	U - - -	70	5 - - -	120
K - - -	25	V - - -	75	6 - - -	125

FIGURE 105-11. Differential switches (MIL-S-9395/32) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
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TABLE III. Codes for combinations of fine settings and tolerance values.

For settings below 20 psid							For settings of 20 psid and above						
	Fine value (in psid)					Tolerance		Fine value (in psid)					Tolerance
	0	1/2	1	1-1/2	2			0	1	2	3	4	
Code - - -	A	B	C	D	E	±1/2 psid	Code - - -	A	B	C	D	E	±1 psid 1/
Code - - -	F	G	H	J	K	±1 psid	Code - - -	F	G	H	J	K	±2 psid 2/
Code - - -	L	M	N	P	Q	±2 psid	Code - - -	L	M	N	P	Q	±3 psid 3/
Code - - -	R	S	T	U	V	±3 psid	Code - - -	R	S	T	U	V	±5 psid
Code - - -	W	X	Y	Z	1	±5 psid	Code - - -	W	X	Y	Z	1	±8 psid
Code - - -	2	3	4	5	6	±8 psid	Code - - -	2	3	4	5	6	±15 psid
Code - - -	7	8	9	0	I	Min or Max	Code - - -	7	8	9	0	I	Min or Max

1/ Not applicable for pressure settings above 33 psid.
2/ Not applicable for pressure settings above 66 psid.
3/ Not applicable for pressure settings above 100 psid.

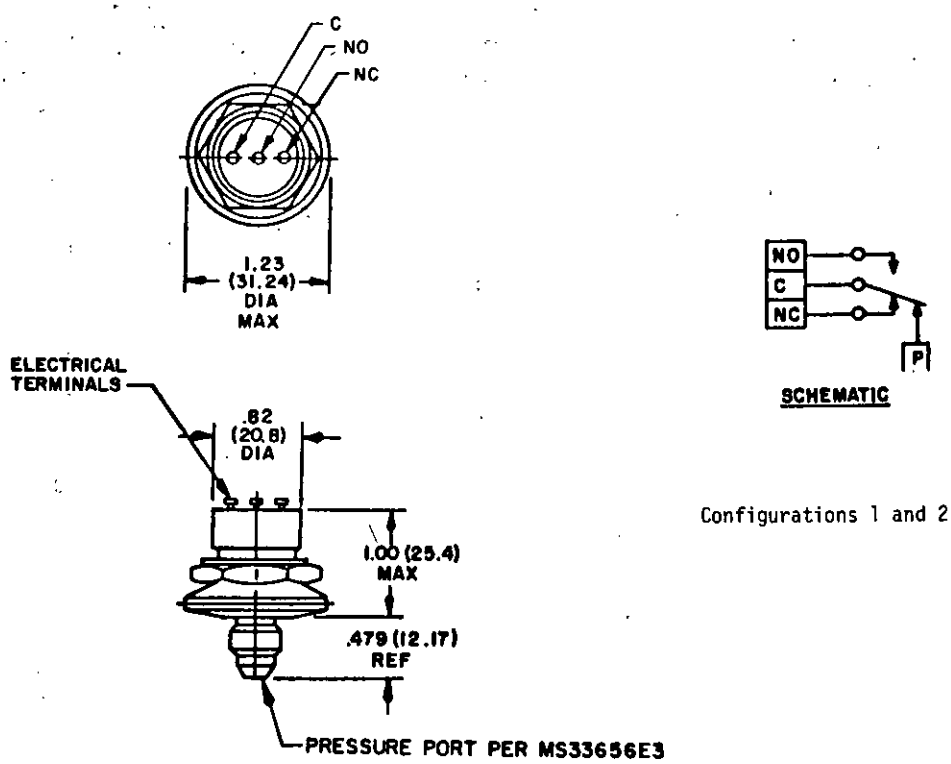
TABLE IV. Codes for differential settings.

Code	Differential value (in psid)	Code	Differential value (in psid)	Code	Differential value (in psid)
A - - -	0	N - - -	6	1 - - -	20
B - - -	1/2	P - - -	7	2 - - -	22
C - - -	1	Q - - -	8	3 - - -	24
D - - -	1-1/2	R - - -	9	4 - - -	26
E - - -	2	S - - -	10	5 - - -	28
F - - -	2-1/2	T - - -	11	6 - - -	30
G - - -	3	U - - -	12	7 - - -	35
H - - -	3-1/2	V - - -	13	8 - - -	40
J - - -	4	W - - -	14	9 - - -	45
K - - -	4-1/2	X - - -	15	0 - - -	50
L - - -	5	Y - - -	16		
M - - -	5-1/2	Z - - -	18		

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

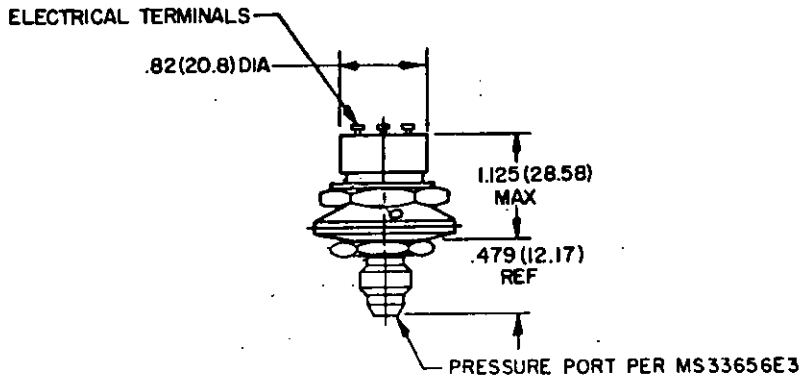
FIGURE 105-11. Differential switches (MIL-S-9395/32) - Continued.

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Configurations 1 and 2

Configuration 1



Configuration 2

FIGURE 105-12. Absolute switches (MIL-S-9395/35).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
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PRESSURE SETTING (WITH TOLERANCE) AND MAXIMUM DIFFERENTIAL VALUES FOR EACH SETTING:

Pressure setting (lb_f/in^2)		Max differential (lb_f/in^2)	Pressure setting (lb_f/in^2)		Max differential (lb_f/in^2)
Value	Tolerance (±)		Value	Tolerance (±)	
1.0 - 10.5	.5	2.5	44.5	1.5	4.6
11.0 - 20.0	.75	2.5	45.0 - 45.5	2.0	4.7
20.5 - 29.5	1.0	3.0	46.0 - 46.5	↓	4.8
30.0 - 30.5	1.0	3.1	47.0 - 47.5	↓	4.9
31.0	1.5	3.2	48.0 - 48.5	↓	5.0
31.5 - 32.0	↓	3.3	49.0 - 49.5	↓	5.1
32.5 - 33.0	↓	3.4	50.0 - 50.5	↓	5.2
33.5 - 34.0	↓	3.5	51.0 - 51.5	↓	5.3
34.5 - 35.0	↓	3.6	52.0 - 52.5	↓	5.4
35.5 - 36.0	↓	3.7	53.0 - 53.5	↓	5.5
36.5 - 37.0	↓	3.8	54.0 - 54.5	↓	5.6
37.5 - 38.0	↓	3.9	55.0 - 55.5	↓	5.7
38.5 - 39.0	↓	4.0	56.0 - 56.5	↓	5.8
39.5 - 40.0	↓	4.1	57.0 - 57.5	↓	5.9
40.5 - 41.0	↓	4.2	58.0 - 58.5	↓	6.0
41.5 - 42.0	↓	4.3	59.0 - 59.5	↓	6.1
42.5 - 43.0	↓	4.4	60.0	2.0	6.2
43.5 - 44.0	1.5	4.5			

PART NUMBER: Consists of "M9395/35" followed by a dash and a three-letter code. The three-letter code in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 18.0 ±.75 lb_f/in^2 with a maximum differential of 2.5 lb_f/in^2.

EXAMPLE:

M9395/35- A G E

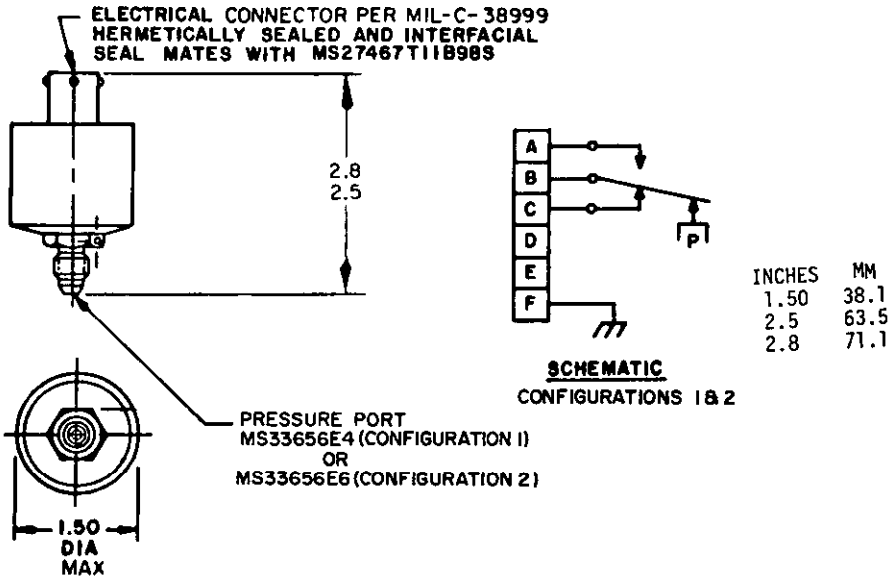
CONFIGURATION AND CONTACT ACTION		
	Configuration	
	1	2
Actuates on increasing pressure	A	C
Actuates on decreasing pressure	B	D

PRESSURE SETTING TO WITHIN 1 lb_f/in^2					
Pressure	Code	Pressure	Code	Pressure	Code
2.0	A	22.0	J	42.0	S
4.5	B	24.5	K	44.5	T
7.0	C	27.0	L	47.0	U
9.5	D	29.5	M	49.5	V
12.0	E	32.0	N	52.0	W
14.5	F	34.5	P	54.5	X
17.0	G	37.0	Q	57.0	Y
19.5	H	39.5	R	59.5	Z

PRESSURE SETTING TO WITHIN .5 lb_f/in^2			
Unit	Code	Unit	Code
-1.0	A	0.5	D
-0.5	B	1.0	E
0	C		

FIGURE 105-12. Absolute switches (MIL-S-9395/35) -Continued.

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PART NUMBER: Consists of "M9395/38" followed by a dash and a five-letter code. The five-letter code in the following example identifies a switch of configuration 1 (low level to 1 ampere resistive at 28 Vdc) which actuates on increasing pressure at 1.5 ±0.5 lb_f/in^2 , and deactuates on decreasing pressure at 0.5 ±0.5 lb_f/in^2.

EXAMPLE:

M9395/38- A A J A G

CONFIGURATION, ELECTRICAL RATING, AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "A" of table II (coarse value of 0) and Code "J" of table III (fine value of 1.5 ±0.5).

LOW-PRESSURE SETTING = Code "A" of table II (coarse value of 0) and Code "G" of table III (fine value of 0.5 ±0.5).

TABLE I. Codes for configurations, electrical ratings, and pressure setting modes.

	Low level to 1 A resistive at 28 Vdc				Pressure setting mode	
	Configuration					
	1		2			
	Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed		
Code - - -	A	D	G	K	At (or max) 1/	At (or min) 1/
Code - - -	B	E	H	L	At (or max) 1/	Differential 2/
Code - - -	C	F	J	M	Differential 2/	At (or min) 1/
	1.5 to 5 A resistive at 28 Vdc				High setting	Low setting
Code - - -	N	R	U	X	At (or max) 1/	At (or min) 1/
Code - - -	P	S	V	Y	At (or max) 1/	Differential 2/
Code - - -	Q	T	W	Z	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.

2/ Setting values are designated by codes from table IV.

FIGURE 105-13. Gage switches (MIL-S-9395/38).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Codes for coarse settings.

Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)
A - -	0	J - -	20	S - -	60	1 - -	100
B - -	2.5	K - -	25	T - -	65	2 - -	105
C - -	5	L - -	30	U - -	70	3 - -	110
D - -	7.5	M - -	35	V - -	75	4 - -	115
E - -	10	N - -	40	W - -	80	5 - -	120
F - -	12.5	P - -	45	X - -	85	6 - -	125
G - -	15	Q - -	50	Y - -	90		
H - -	17.5	R - -	55	Z - -	95		

TABLE III. Codes for combinations of fine settings and tolerance values.

	Settings below 20 lb_f/in^2					Tolerance (lb_f/in^2)
	0	.5	1	1.5	2	
Code - - -	A	B	C	D	E	±0.25
Code - - -	F	G	H	J	K	±0.5
Code - - -	L	M	N	P	Q	±1.0
Code - - -	R	S	T	U	V	±1.5
Code - - -	W	X	Y	Z	1	±2.0
Code - - -	2	3	4	5	6	±2.5
Code - - -	7	8	9	0	-	Min or Max
	Settings of 20 lb_f/in^2 and above					Tolerance (lb_f/in^2)
	0	1	2	3	4	
Code - - -	A	B	C	D	E	±1.0 1/
Code - - -	F	G	H	J	K	±2.0 2/
Code - - -	L	M	N	P	Q	±3.0 3/
Code - - -	R	S	T	U	V	±4.0
Code - - -	W	X	Y	Z	1	±5.0
Code - - -	2	3	4	5	6	±6.0
Code - - -	7	8	9	0	-	Min or Max

1/ Not applicable for pressure settings above 33 lb_f/in^2.
2/ Not applicable for pressure settings above 66 lb_f/in^2.
3/ Not applicable for pressure settings above 100 lb_f/in^2.

TABLE IV. Codes for differential settings. 1/

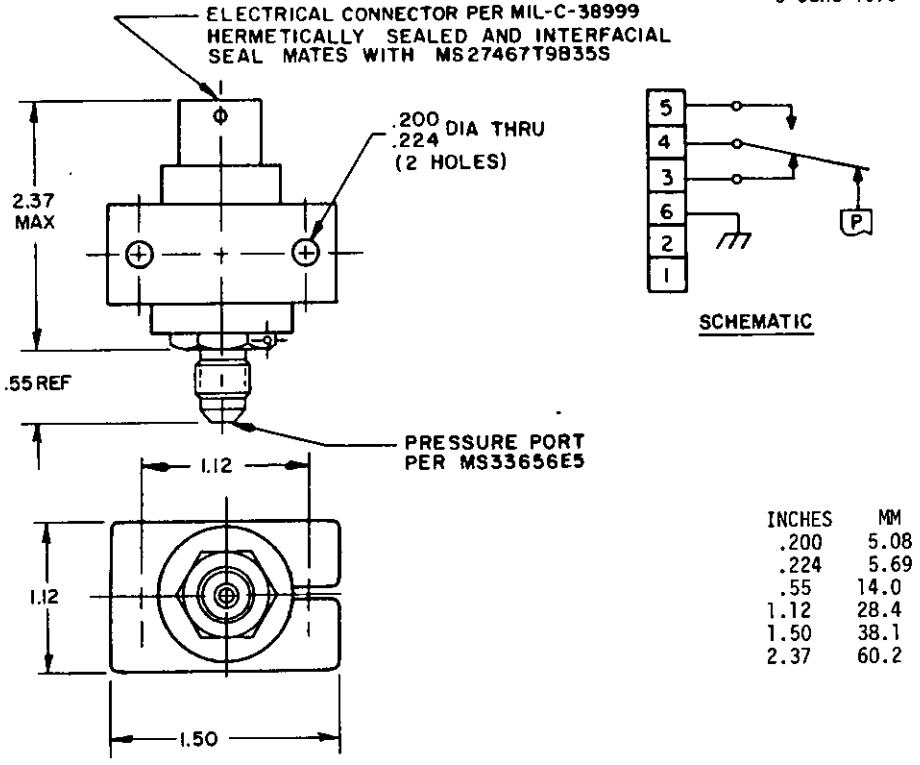
Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	N - - -	6	1 - - -	20
B - - -	.5	P - - -	7	2 - - -	22
C - - -	1	Q - - -	8	3 - - -	24
D - - -	1.5	R - - -	9	4 - - -	26
E - - -	2	S - - -	10	5 - - -	28
F - - -	2.5	T - - -	11	6 - - -	30
G - - -	3	U - - -	12	7 - - -	35
H - - -	3.5	V - - -	13	8 - - -	40
J - - -	4	W - - -	14	9 - - -	45
K - - -	4.5	X - - -	15	0 - - -	50
L - - -	5	Y - - -	16		
M - - -	5.5	Z - - -	18		

1/ Differential settings require two codes, minimum differential and maximum differential.

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

FIGURE 105-13. Gage switches (MIL-S-9395/38) -Continued.

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INCHES	MM
.200	5.08
.224	5.69
.55	14.0
1.12	28.4
1.50	38.1
2.37	60.2

CONFIGURATION 1

PART NUMBER: Consists of "M9395/39" followed by a dash and a five-letter code. The five-letter code in the following example identifies a switch of configuration 1 (low level to 1 ampere resistive at 28 Vdc) which actuates on increasing pressure at 1.5 ±0.5 lb_f/in^2 and deactuates on decreasing pressure at 0.5 ±0.5 lb_f/in^2.

EXAMPLE:

M9395/39- A A J A G

CONFIGURATION, ELECTRICAL RATING, AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "A" of table II (coarse value of 0) and Code "J" of table III (fine value of 1.5 ±0.5).

LOW-PRESSURE SETTING = Code "A" of table II (coarse value of 0) and Code "G" of table III (fine value of 0.5 ±0.5).

TABLE I. Codes for configurations, electrical ratings, and pressure setting modes.

	Low level to 1 A resistive		1.5 to 5 A resistive		Pressure setting mode	
	Configuration 1					
	Electrical chamber					
	Hermetic	Unsealed	Hermetic	Unsealed	High pressure	Low pressure
Code - -	A	D	G	K	At (or max) 1/	At (or min) 1/
Code - -	B	E	H	L	At (or max) 1/	Differential 2/
Code - -	C	F	J	M	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

FIGURE 105-14. Pressure switches (MIL-S-9395/39).

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TABLE II. Codes for coarse settings.

Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)
A - -	0	J - -	20	S - -	60	1 - -	100
B - -	2.5	K - -	25	T - -	65	2 - -	105
C - -	5	L - -	30	U - -	70	3 - -	110
D - -	7.5	M - -	35	V - -	75	4 - -	115
E - -	10	N - -	40	W - -	80	5 - -	120
F - -	12.5	P - -	45	X - -	85	6 - -	125
G - -	15	Q - -	50	Y - -	90		
H - -	17.5	R - -	55	Z - -	95		

TABLE III. Codes for combinations of fine settings and tolerance values.

	Settings below 20 lb_f/in^2					Tolerance (lb_f/in^2)
	0	.5	1	1.5	2	
Code - - -	A	B	C	D	E	±0.25
Code - - -	F	G	H	J	K	±0.5
Code - - -	L	M	N	P	Q	±1.0
Code - - -	R	S	T	U	V	±1.5
Code - - -	W	X	Y	Z	1	±2.0
Code - - -	2	3	4	5	6	±2.5
Code - - -	7	8	9	0	-	Min or Max
	Settings of 20 lb_f/in^2 and above					Tolerance (lb_f/in^2)
	0	1	2	3	4	
Code - - -	A	B	C	D	E	±1.0 1/
Code - - -	F	G	H	J	K	±2.0 2/
Code - - -	L	M	N	P	Q	±3.0 3/
Code - - -	R	S	T	U	V	±4.0
Code - - -	W	X	Y	Z	1	±5.0
Code - - -	2	3	4	5	6	±6.0
Code - - -	7	8	9	0	-	Min or Max

1/ Not applicable for pressure settings above 33 lb_f/in^2.
2/ Not applicable for pressure settings above 66 lb_f/in^2.
3/ Not applicable for pressure settings above 100 lb_f/in^2.

TABLE IV. Codes for differential settings. 1/

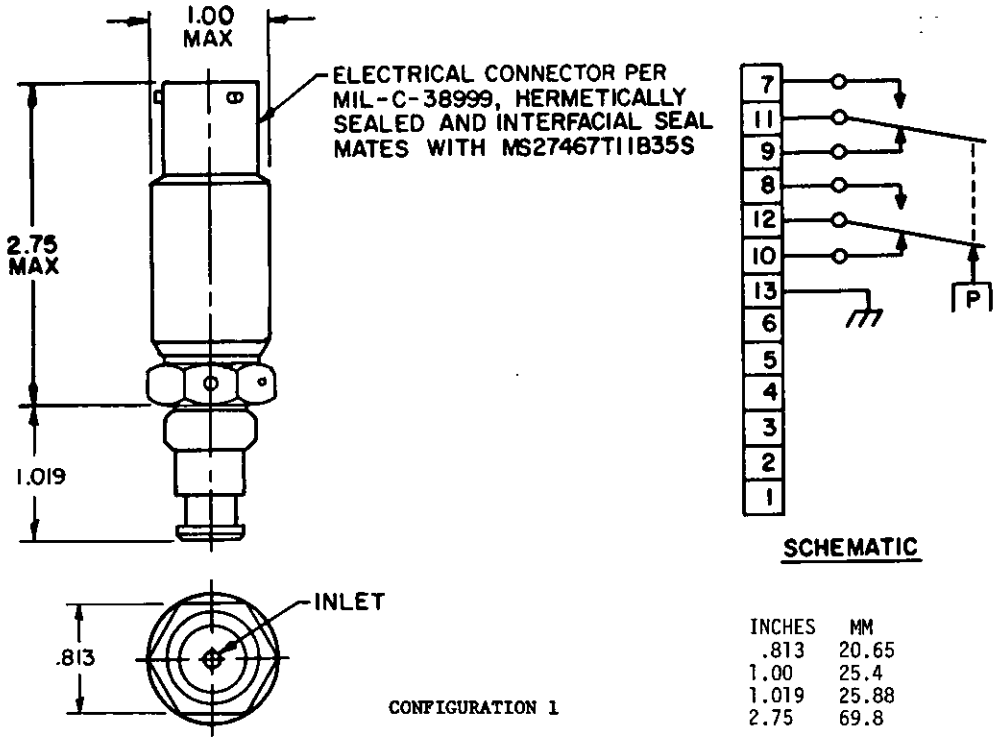
Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	N - - -	6	1 - - -	20
B - - -	.5	P - - -	7	2 - - -	22
C - - -	1	Q - - -	8	3 - - -	24
D - - -	1.5	R - - -	9	4 - - -	26
E - - -	2	S - - -	10	5 - - -	28
F - - -	2.5	T - - -	11	6 - - -	30
G - - -	3	U - - -	12	7 - - -	35
H - - -	3.5	V - - -	13	8 - - -	40
J - - -	4	W - - -	14	9 - - -	45
K - - -	4.5	X - - -	15	0 - - -	50
L - - -	5	Y - - -	16		
M - - -	5.5	Z - - -	18		

1/ Differential settings require two codes, minimum differential and maximum differential.

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

FIGURE 105-14. Pressure switches (MIL-S-9395/39) -Continued.

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INCHES	MM
.813	20.65
1.00	25.4
1.019	25.88
2.75	69.8

PART NUMBER: Consists of "M9395/40" followed by a dash and a five-letter code. The five-letter code in the following example identifies a switch of configuration 1 (low level to 1 ampere resistive) which actuates on increasing pressure at 300 ±25 lb_f/in^2 and deactuates on decreasing pressure at 150 ±25 lb_f/in^2.

EXAMPLE:

M9395/40- A C A A C

CONFIGURATION, ELECTRICAL RATING, AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "C" of table II (300) and Code "A" of table III (0, ±25).

LOW-PRESSURE SETTING = Code "A" of table II (100) and Code "C" of table III (+50, ±25).

TABLE I. Codes for configurations, electrical ratings, and pressure setting modes.

	Configuration 1		Pressure setting mode	
	Low level to 1A resistive	1.5 to 5A resistive	High setting	Low setting
Code - - -	A	D	At (or max) 1/	At (or min) 1/
Code - - -	B	E	At (or max) 1/	Differential 2/
Code - - -	C	F	Differential 2/	At (or min) 1/

1/ Setting values are designated by characters from tables II and III.
2/ Setting values are designated by characters from table IV.

FIGURE 105-15. Gage switches (MIL-S-9395/40).

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TABLE II. Codes for pressure settings to within 100 lb_f/in^2.

Code	Pressure (in lb_f/in^2)	Code	Pressure (in lb_f/in^2)	Code	Pressure (in lb_f/in^2)	Code	Pressure (in lb_f/in^2)
A - - -	100	J - - -	900	S - - -	1700	1 - - -	2500
B - - -	200	K - - -	1000	T - - -	1800	2 - - -	2600
C - - -	300	L - - -	1100	U - - -	1900	3 - - -	2700
D - - -	400	M - - -	1200	V - - -	2000	4 - - -	2800
E - - -	500	N - - -	1300	W - - -	2100	5 - - -	2900
F - - -	600	P - - -	1400	X - - -	2200	6 - - -	3000
G - - -	700	Q - - -	1500	Y - - -	2300		
H - - -	800	R - - -	1600	Z - - -	2400		

TABLE III. Codes for pressure settings to within 25 lb_f/in^2 and tolerances.

	Unit				Tolerance (in lb_f/in^2)
	0	+25	+50	+75	
Code - - -	A	B	C	D	±25
Code - - -	E	F	G	H	±50
Code - - -	J	K	L	M	±75
Code - - -	N	P	Q	R	±100
Code - - -	S	T	U	V	±150
Code - - -	W	X	Y	Z	±200
Code - - -	1	2	3	4	±300
Code - - -	5	6	7	8	±400
Code - - -	9	0	I	-	Min or Max

TABLE IV. Codes for differential pressure settings. 1/

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	M - - -	275
B - - -	25	N - - -	300
C - - -	50	P - - -	325
D - - -	75	Q - - -	350
E - - -	100	R - - -	375
F - - -	125	S - - -	400
G - - -	150	T - - -	425
H - - -	175	U - - -	450
J - - -	200	V - - -	475
K - - -	225	W - - -	500
L - - -	250		

1/ Differential settings require two codes, minimum differential and maximum differential.

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

FIGURE 105-15. Gage switches (MIL-S-9395/40) -Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
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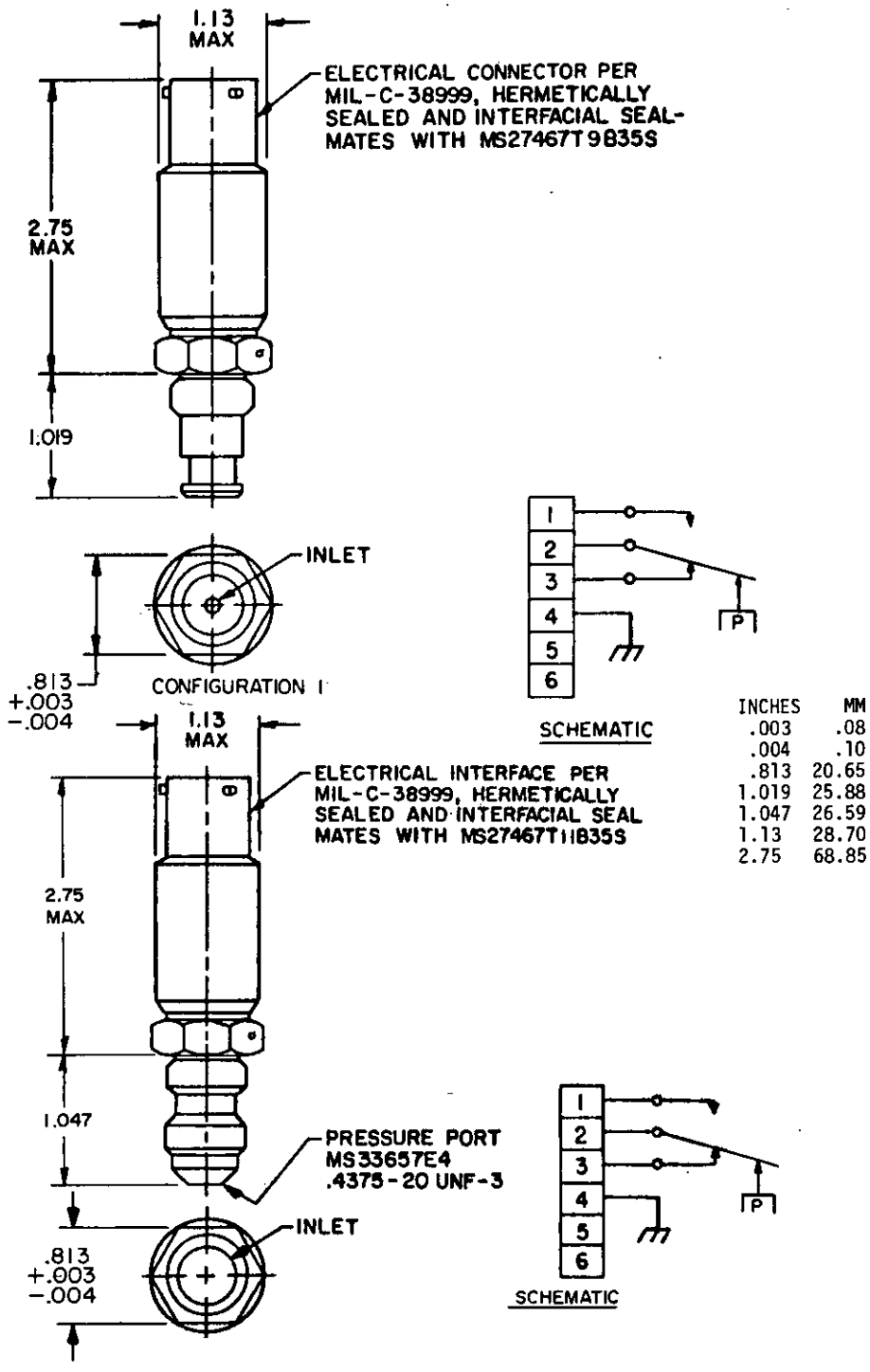


FIGURE 105-16. Pressure switches (MIL-S-9395/41).

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PART NUMBER: Consists of "M9395/41" followed by a dash and a five-letter code. The five-letter code in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 1.5 ±0.5 lb_f/in^2 and deactuates on decreasing pressure at 0.5 ±0.5 lb_f/in^2.

EXAMPLE:

M9395/41- A A J A G

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I.

HIGH-PRESSURE SETTING = Code "A" of table II (coarse value of 0) and Code "J" of table III (fine value of 1.5 ±0.5).

LOW-PRESSURE SETTING = Code "A" of table II (coarse value of 0) and Code "G" of table III (fine value of 0.5 ±0.5).

TABLE I. Codes for combinations of configurations and pressure setting modes.

	Configuration		Pressure setting mode	
	1	2	High setting	Low setting
Code - - -	A	D	At (or max) 1/	At (or min) 1/
Code - - -	B	E	At (or max) 1/	Differential 2/
Code - - -	C	F	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

TABLE II. Codes for coarse settings.

Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)	Code	Coarse value (lb_f/in^2)
A - -	0	J - -	20	S - -	60	1 - -	100
B - -	2.5	K - -	25	T - -	65	2 - -	105
C - -	5	L - -	30	U - -	70	3 - -	110
D - -	7.5	M - -	35	V - -	75	4 - -	115
E - -	10	N - -	40	W - -	80	5 - -	120
F - -	12.5	P - -	45	X - -	85	6 - -	125
G - -	15	Q - -	50	Y - -	90		
H - -	17.5	R - -	55	Z - -	95		

FIGURE 105-16. Pressure switches (MIL-S-9395/41) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
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TABLE III. Codes for combinations of fine settings and tolerance values.

	Settings below 20 lb_f/in^2					Tolerance (lb_f/in^2)
	0	.5	1	1.5	2	
Code - - -	A	B	C	D	E	±0.25
Code - - -	F	G	H	J	K	±0.5
Code - - -	L	M	N	P	Q	±1.0
Code - - -	R	S	T	U	V	±1.5
Code - - -	W	X	Y	Z	1	±2.0
Code - - -	2	3	4	5	6	±2.5
Code - - -	7	8	9	0	-	Min or Max
	Settings of 20 lb_f/in^2 and above					Tolerance (lb_f/in^2)
	0	1	2	3	4	
Code - - -	A	B	C	D	E	±1.0 1/
Code - - -	F	G	H	J	K	±2.0 2/
Code - - -	L	M	N	P	Q	±3.0 3/
Code - - -	R	S	T	U	V	±4.0
Code - - -	W	X	Y	Z	1	±5.0
Code - - -	2	3	4	5	6	±6.0
Code - - -	7	8	9	0	-	Min or Max

1/ Not applicable for pressure settings above 33 lb_f/in^2.
2/ Not applicable for pressure settings above 66 lb_f/in^2.
3/ Not applicable for pressure settings above 100 lb_f/in^2.

TABLE IV. Codes for differential settings. 1/

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A - - -	0	N - - -	6	1 - - -	20
B - - -	.5	P - - -	7	2 - - -	22
C - - -	1	Q - - -	8	3 - - -	24
D - - -	1.5	R - - -	9	4 - - -	26
E - - -	2	S - - -	10	5 - - -	28
F - - -	2.5	T - - -	11	6 - - -	30
G - - -	3	U - - -	12	7 - - -	35
H - - -	3.5	V - - -	13	8 - - -	40
J - - -	4	W - - -	14	9 - - -	45
K - - -	4.5	X - - -	15	0 - - -	50
L - - -	5	Y - - -	16		
M - - -	5.5	Z - - -	18		

1/ Differential settings require two codes, minimum differential and maximum differential.

NOTE: Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number.

FIGURE 105-16. Pressure switches (MIL-S-9395/41) -Continued.

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SECTION 112

SWITCHES, TOGGLE
(Applicable specifications: MIL-S-3950, MIL-S-8834, and MIL-S-8805)

FIGURE AND SPEC NUMBER	MS NUMBER (see NOTE)	SWITCH CHARACTERISTICS			PAGE
		Dash No. for:		Circuit characteristics	
		3 Position	2 Position		
112-1 - MIL-S-3950	MS24523	21,27,31	22,23,26,29,30	One pole	112.3
112-2 - MIL-S-3950	MS24524	21,27,31	22,23,26,29,30	Two pole	112.4
	MS24525	21,27,31	22,23,26,29,30	Four pole	
	MS27406	1,2,3	---	Four pole	
	MS27407	1,2,3	---	Two pole	
112-3 - MIL-S-3950	* MS24658	21,27,31	22,23,26,28,29, 30,32	One pole	112.5
112-4 - MIL-S-3950	* MS24659	21,27,31	22,23,26,28,29, 30,32	Two pole	112.6
	* MS24660	21,27,31	22,23,26,28,29, 30,32	Four pole	
	* MS27408	1,2,3	---	Two pole	
	* MS27409	1,2,3	---	Four pole	
112-5 - MIL-S-3950	MS27722	21,27,31	22,23,26,28,29, 30,32	One pole	112.7
	MS27784	21,27,31	22,23,26,28,29, 30,32	One pole	
112-6 - MIL-S-3950	MS27723	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.8
	MS27724	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-7 - MIL-S-3950	MS27785	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.9
	MS27786	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-8 - MIL-S-3950	* MS27782	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.10
	* MS27783	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-9 - MIL-S-3950	* MS27788	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.11
	* MS27789	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-10 - MIL-S-3950	* MS27781	21,27,31	22,23,26,28,29, 30,32	One pole	112.12
	* MS27787	21,27,31	22,23,26,28,29, 30,32	One pole	
112-11 - MIL-S-3950	MS27734	21,27,31	22,23,26,28,29, 30,32	One pole	112.13
112-12 - MIL-S-3950	MS27735	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.14
	MS27736	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-13 - MIL-S-3950	* MS27737	21,27,31	22,23,26,28,29, 30,32	One pole	112.15
112-14 - MIL-S-3950	* MS27738	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.16
	* MS27739	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-15 - MIL-S-8834	MS14001	212,272,312	222,232,242, 262,282,292,302	One pole	112.17
	MS14002	212,272,312	222,232,242, 262,282,292,302	Two pole	
	MS14003	212,272,312	222,232,242, 262,282,292,302	Four pole	

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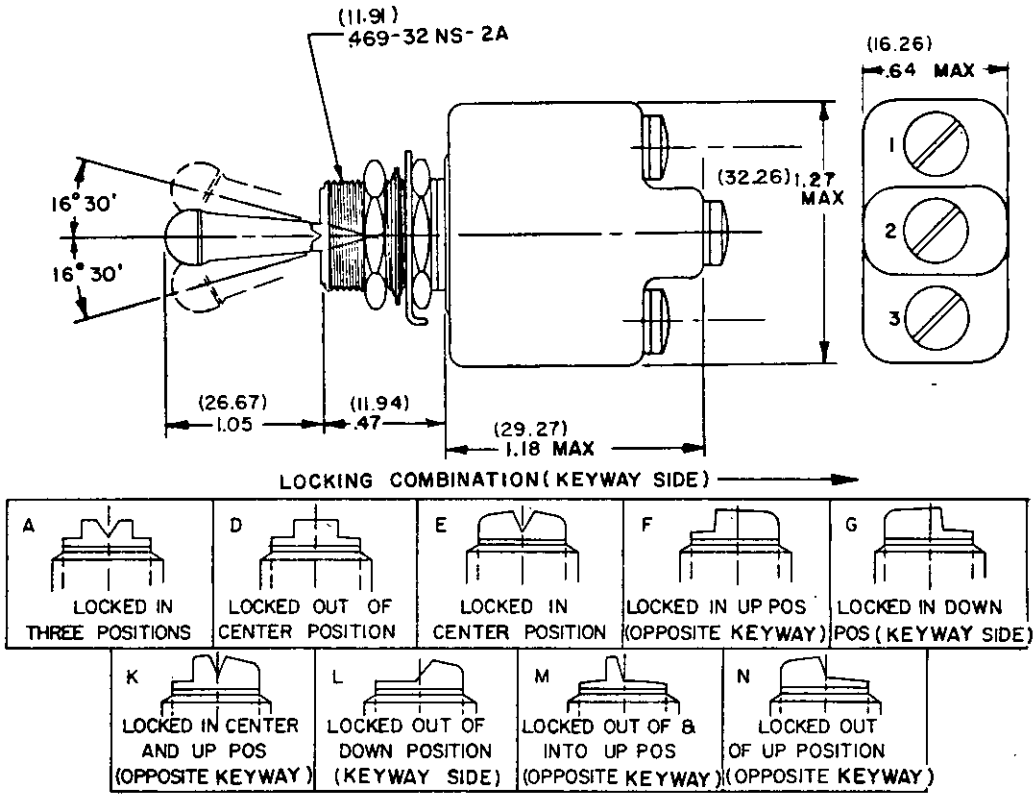
FIGURE AND SPEC NUMBER	MS NUMBER (see NOTE)	SWITCH CHARACTERISTICS Dash No. for: 3 Position	2 Position	Circuit characteristics	PAGE
112-16 - MIL-S-8834	* MS21026	211,271,311	221,231,241,281	One pole	112.18
	* MS21027	211,311,371,711, 771,811,871	221,231,241,381,721, 731,741,781,881	Two pole	
112-17 - MIL-S-8834	MS21346	211,271,311	221,231,241,281	One pole	112.19
	MS21347	211,271,311,711, 771,811	221,231,241,281,721, 731,741,781	Two pole	
112-18 - MIL-S-8834	MS21350	---	321	One pole	112.20
	MS21351	---	321	One pole	
112-19 - MIL-S-8834	MS21352	331,341,351	321	Two pole	112.21
	MS21353	331,341,351,831, 841,851	321,821	Two pole	
112-20 - MIL-S-8834	MS21354	211,271,311	221,231,241,281	One pole	112.22
	MS21355	211,271,311	221,231,241,281	Two pole	
112-21 - MIL-S-8834	MS21356	211,271,311	221,231,241,281	One pole	112.23
	MS21357	211,271,311	221,231,241,281	Two pole	
112-22 - MIL-S-8834	* MS24612	212,272,312	222,232,242,262, 282,292,302	One pole	112.24
	* MS24613	212,272,312	222,232,242,262, 282,292,302	Two pole	
	* MS24614	212,272,312	222,232,242,262, 282,292,302	Four pole	
112-23 - MIL-S-8834	MS24655	211,271,311	221,231,241,281	One pole	112.25
	MS24656	211,271,311	221,231,241,281	Two pole	
112-24 - MIL-S-8834	MS25306	212,272,312	222,232,242,262, 282,292,302	One pole	112.26
	MS25307	212,272,312	222,232,242,262, 282,292,302	Two pole	
	MS25308	212,272,312	222,232,242,262, 282,292,302	Four pole	
112-25 - MIL-S-8834	MS90310	211,271,311	221,231,241,281	One pole	112.27
	MS90311	211,271,311,711, 771,811	221,231,241,281, 721,731,741,781	Two pole	
112-26 - MIL-S-8805/26	*	005,006,007, 008,009,010	001,003	Two pole	112.28
		011,012,013, 014	---	Three pole	
		---	002,004	Four pole	
112-27 - MIL-S-8805/92	*	005,006,007, 008,009,010, 011,012,013, 014,015,016, 017,018	001	Two pole	112.30
		019	---	Three pole	
		---	002,003,004	Four pole	
112-28 - MIL-S-8805/93	*	001 thru 007 008 thru 014 015 thru 021 022 thru 028	--- --- --- ---	Two pole Four pole Six pole Eight pole	112.32
112-29 - MIL-S-8805/103	*	001,005,009, 012,014,016, 017	---	Four pole	112.34
		---	004,010	Six pole	
		002,011,013, 015,018	---	Eight pole	

NOTE: The part number consists of the MS number followed by the dash number; however, an asterisk (*) indicates that the "MS" part number will require an additional letter, or an "M" part number is applicable, as shown in the figure.

Supersedes page 112.2
of 20 June 1977

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DASH NO.	CURRENT CAPACITY (amperes)					
	28 Vdc Load			115 Vac, 60 & 400 Hz Load		
	Lamp	Resistive	Inductive	Lamp	Resistive	Inductive
-21, -22, & -23	5	20	15	3	15	10
-26, -27, -28, -29, -30, -31, & -32	4	15	10	2	15	7



Dash No.	Available locking combinations	1/ Circuit with toggle lever in			Dash No.	Available locking combinations	1/ Circuit with toggle lever in		
		Opposite keyway side	Center	Keyway side			Opposite keyway side	Center	Keyway side
-21	A	ON	OFF	ON	-28	E	NONE	OFF	MOM ON
-22	D,F,G	ON	NONE	OFF	-29	F	ON	NONE	MOM OFF
-23	D,F,G	ON	NONE	ON	-30	F	OFF	NONE	MOM ON
-26	F	ON	NONE	MOM ON	-31	E,F,K,L,M,N	ON	OFF	MOM ON
-27	E,L,N	MOM ON	OFF	MOM ON	-32	E	MOM ON	ON	NONE

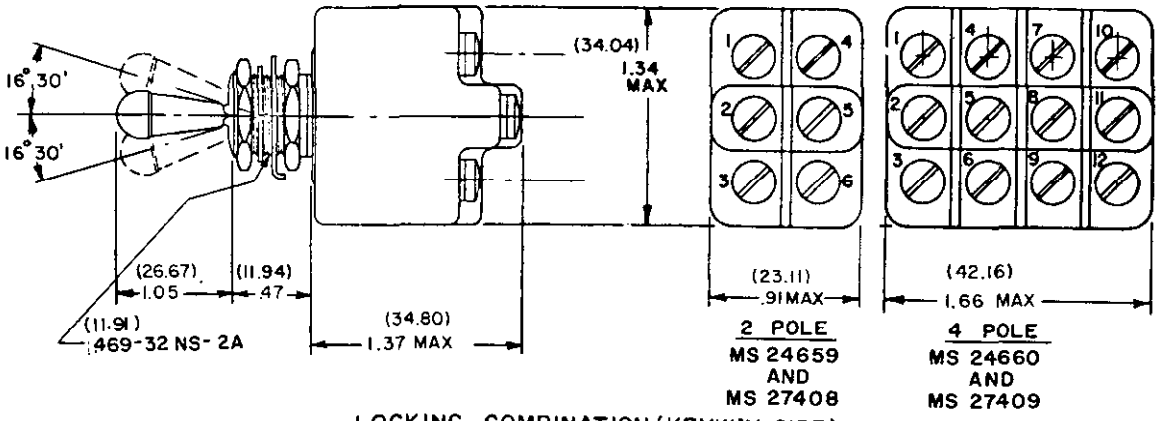
1/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of "MS24658" followed by the dash number and locking combination letter from the tabulation.

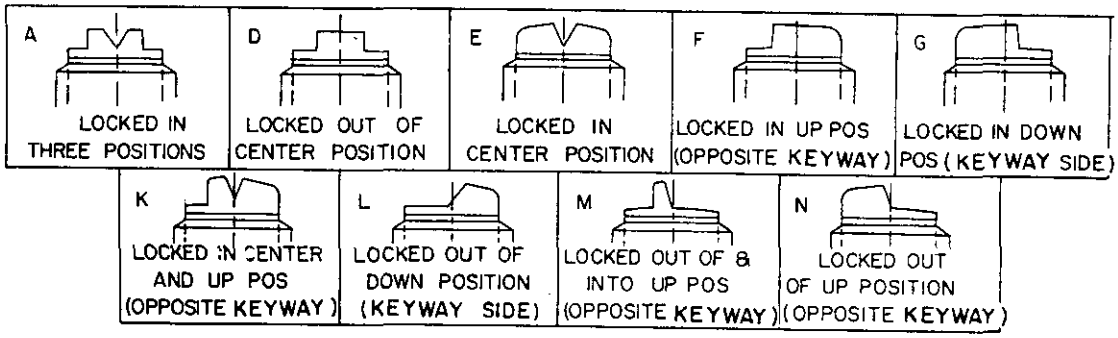
FIGURE 112-3. MS24658 toggle switches, environmentally sealed, lever lock (MIL-S-3950).

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DASH NO.	CURRENT CAPACITY (amperes)							
	28 Vdc					115 Vac, 60 & 400 Hz		
	Load					Load		
	Lamp		Resistive	Inductive		Lamp	Resistive	Inductive
	2 Pole	4 Pole		2 Pole	4 Pole			
-1, -21, -22, & -23	7	5	20	15	12	4	15	15
-2, -3, -26, -27, -28, -29, -30, -31, & -32	5	4	18	10	10	2	11	8



LOCKING COMBINATION (KEYWAY SIDE) →



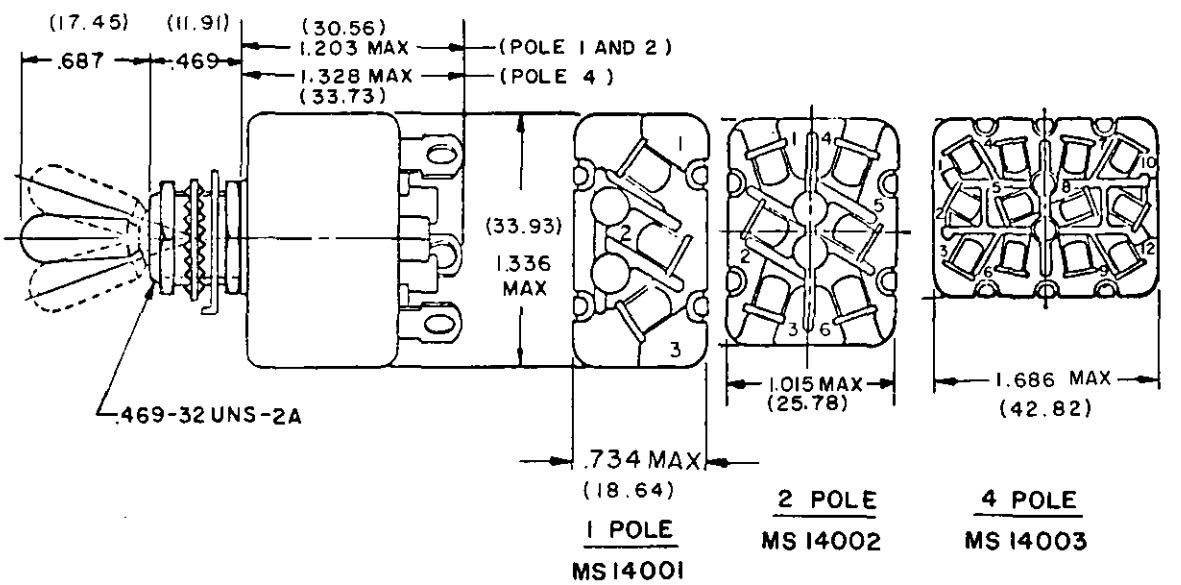
1/ Dash No.	Available locking combination	2/ Circuit with toggle lever in			1/ Dash No.	Available locking combination	2/ Circuit with toggle lever in		
		Opposite keyway side	Center	Keyway side			Opposite keyway side	Center	Keyway side
-1	A	ON	ON	ON	-27	E,L,N	MOM ON	OFF	MOM ON
-2	E,F,K,L,M,N	ON	ON	MOM ON	-28	E	NONE	OFF	MOM ON
-3	E,L,N	MOM ON	ON	MOM ON	-29	F	ON	NONE	MOM OFF
-21	A	ON	OFF	ON	-30	F	OFF	NONE	MOM ON
-22	D,F,G	ON	NONE	OFF	-31	E,F,K,L,M,N	ON	OFF	MOM ON
-23	D,F,G	ON	NONE	ON	-32	E	MOM ON	ON	NONE
-26	F	ON	NONE	MOM ON					

1/ -1, -2, and -3 applicable to MS27408 and MS27409; -21 through -32 applicable to MS24659 and MS24660.
2/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of applicable MS number followed by the dash number and locking combination letter as shown in the tabulation.

FIGURE 112-4. MS24659, MS24660, MS27408, and MS27409 toggle switches, environmentally sealed, lever lock (MIL-S-3950).

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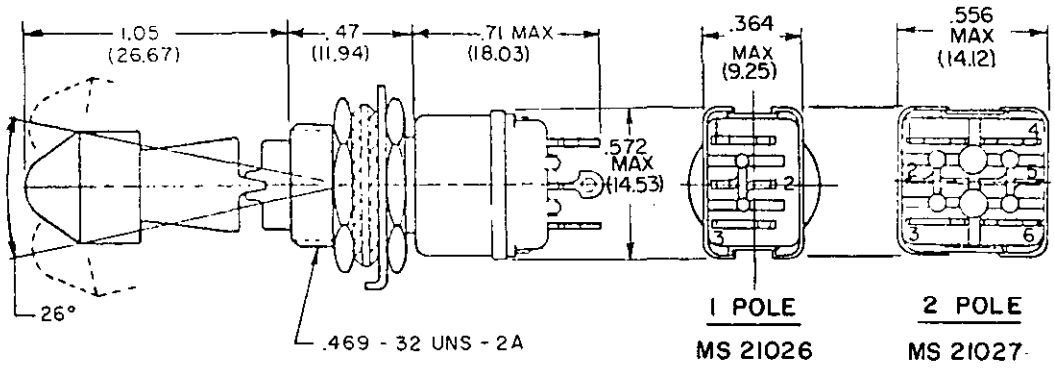
Dash number	1/ Circuit with toggle lever in			Current capacity (amperes)					
	Opposite keyway side	Center	Keyway side	28 Vdc and 115 Vac, 400 Hz Load			115 Vac, 60 Hz Load		
				Lamp	Resistive	Inductive	Lamp	Resistive	Inductive
-212	ON	OFF	ON	7	25	15	7	20	15
-222	ON	NONE	OFF						
-232	ON	NONE	ON						
-242	ON	OFF	NONE						
-262	ON	NONE	MOM-ON						
-272	MOM-ON	OFF	MOM-ON						
-282	NONE	OFF	MOM-ON						
-292	ON	NONE	MOM-OFF						
-302	OFF	NONE	MOM-ON						
-312	ON	OFF	MOM-ON						

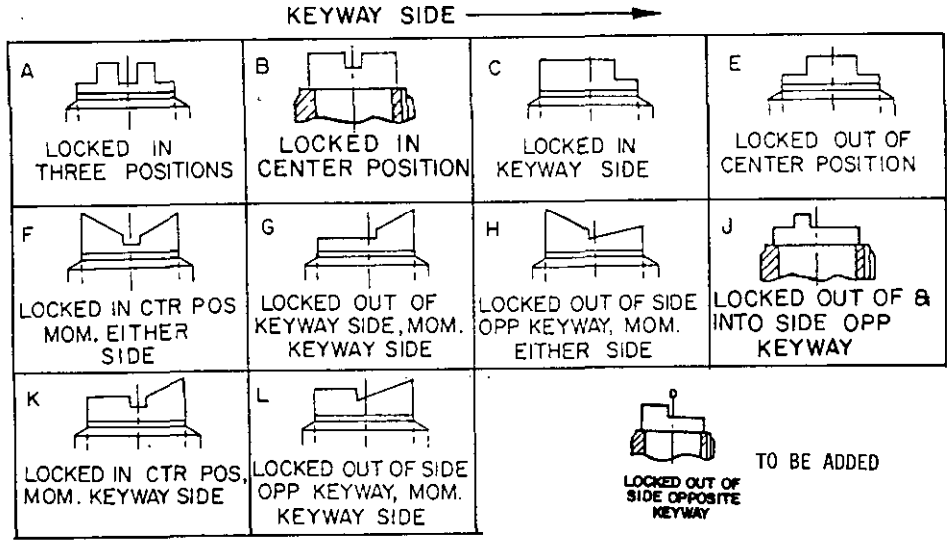
1/ *Applicable terminal numbers are marked on the switch.*

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

FIGURE 112-15. MS14001, MS14402, and MS14003 toggle switches, positive break, environmentally sealed, solder lug, .469 mounting bushing (MIL-S-8834).

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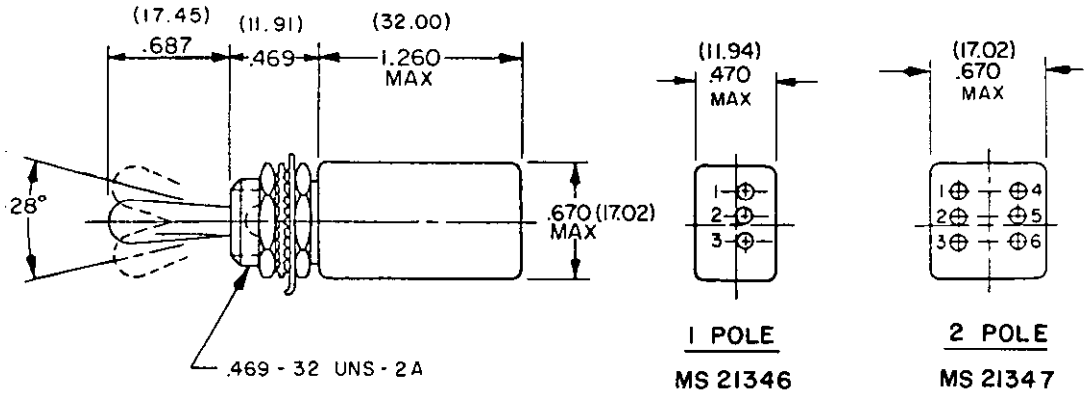
MS dash No. 1/		Locking comb.	Circuit with toggle lever in 2/			Current capacity (in amperes)						Life low current level switching 5 mV
						Resistive load			Inductive load			
			Opposite keyway side	Center position	Keyway side	28 Vdc	115 Vac 60 Hz	115 Vac 400 Hz	28 Vdc	115 Vac 60 Hz	115 Vac 400 Hz	
-711	-211	A,B,D	ON	OFF	ON							
-721	-221	C,E	ON	NONE	OFF							
-731	-231	C,E	ON	NONE	ON							
-741	-241	J,B	ON	OFF	NONE							
-771	-271 *	F,H	MOM-ON	OFF	MOM-ON	5	2	3	1	1	2	.25 µA
-781	-281 *	K	NONE	OFF	MOM-ON							
-811	-311	G,K,L	ON	OFF	MOM-ON							
-871	-371	F,H	MOM-ON	OFF	MOM-ON							
-881	-381	K	NONE	OFF	MOM-ON							

1/ 700 and 800 series have silicone seals (applicable to MS21027 only); 200 and 300 series have mechanical seals. * indicates switch inactive for design (MS21027 only).
2/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the locking combination letter and the three-digit number as shown in the tabulation.

FIGURE 112-16. MS21026 and MS21027 toggle switches, positive break, lever lock, miniature, toggle sealed, solder lug, .469 mounting bushing (MIL-S-8834).

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MS dash No. 1/		2/ Circuit with toggle lever in these positions:			Current capacity (in amperes)						Life low current level switching 5 mV
					Resistive load			Inductive load			
						115 Vac			115 Vac		
		Opposite keyway side	Center position	Keyway side	28 Vdc	60 Hz	400 Hz	28 Vdc	60 Hz	400 Hz	
-711	-211	ON	OFF	ON							
-721	-221	ON	NONE	OFF							
-731	-231	ON	NONE	ON							
-741	-241	ON	OFF	NONE	5	2	3	1	1	2	25 µA
-771	-271	MOM-ON	OFF	MOM-ON							
-781	-281	NONE	OFF	MOM-ON							
-811	-311	ON	OFF	MOM-ON							

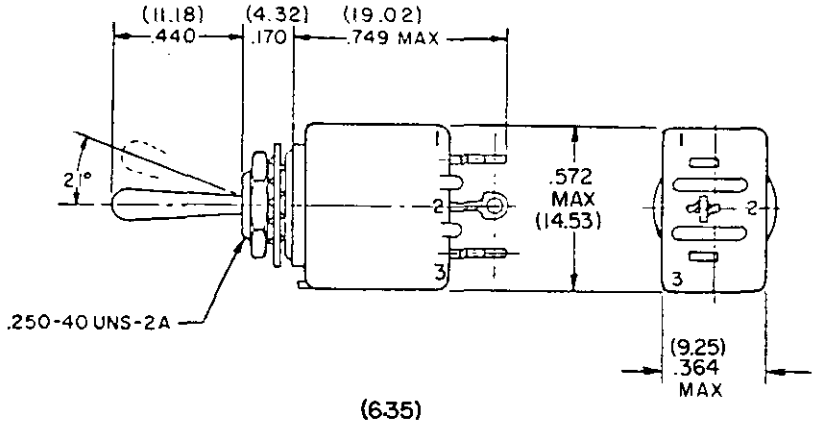
1/ 700 and 800 series have silicone seals (applicable to MS21347 only); 200 and 300 series have mechanical seals (applicable to MS21346 and MS21347).
2/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

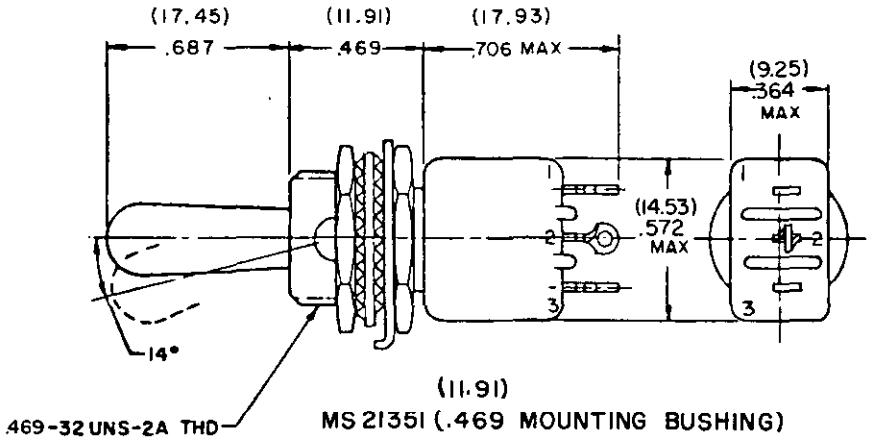
FIGURE 112-17. MS21346 and MS21347 toggle switches, positive break, miniature, toggle sealed, integrated wire terminals, .469 mounting bushing (MIL-S-8834).

Supersedes page 112.19
of 20 June 1977

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(6.35)
MS 21350 (.250 MOUNTING BUSHING)



(11.91)
MS 21351 (.469 MOUNTING BUSHING)

Dash number	1/ Circuit with toggle lever in			Current capacity (amperes)						Life low current level switching 5 mV
	Opposite keyway side	Center	Keyway side	Resistive load			Inductive load			
				28 Vdc	115 Vac		28 Vdc	115 Vac		
					60 Hz	400 Hz		60 Hz	400 Hz	
										(μA)
-321	NONE	ON	MOM-ON	5	2	3	1	1	2	25

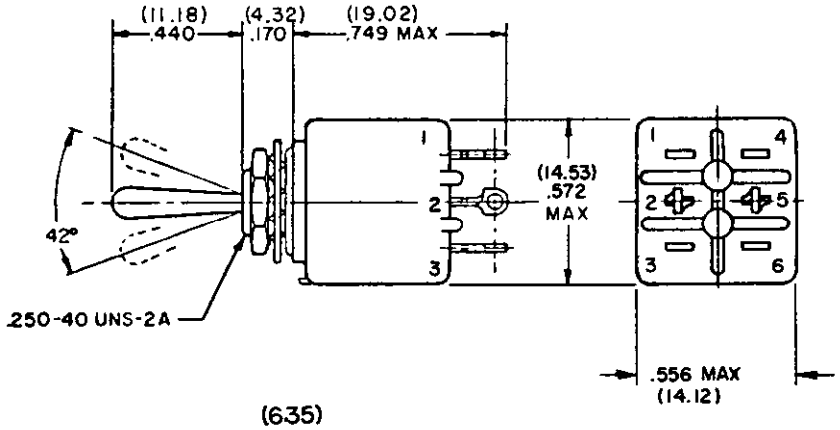
1/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

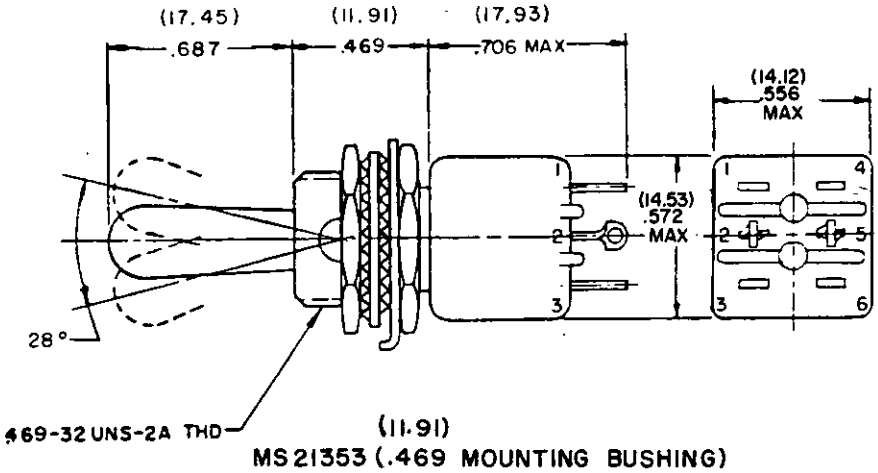
FIGURE 112-18. MS21350 and MS21351 toggle switches, positive break, special circuit, miniature, toggle sealed, solder lug, single pole (MIL-S-8834).

Supersedes page 112.20
of 23 March 1976

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(635)
MS 21352 (.250 MOUNTING BUSHING)



(11.91)
MS 21353 (.469 MOUNTING BUSHING)

MS dash No. 1/		2/ Circuit with toggle lever in			Current capacity (amperes)						Life low current level switching 5 mV
					Resistive load			Inductive load			
		Opposite locking tab side	Center	Locking tab side	28 Vdc	115 Vac 60 Hz	115 Vac 400 Hz	28 Vdc	115 Vac 60 Hz	115 Vac 400 Hz	
											(μA)
-821	-321	NONE	ON	MOM-ON	5	2	3	1	1	2	25
-831	-331	ON	ON	ON.							
-841	-341	MOM-ON	ON	MOM-ON							
-851	-351	ON	ON	MOM-ON							

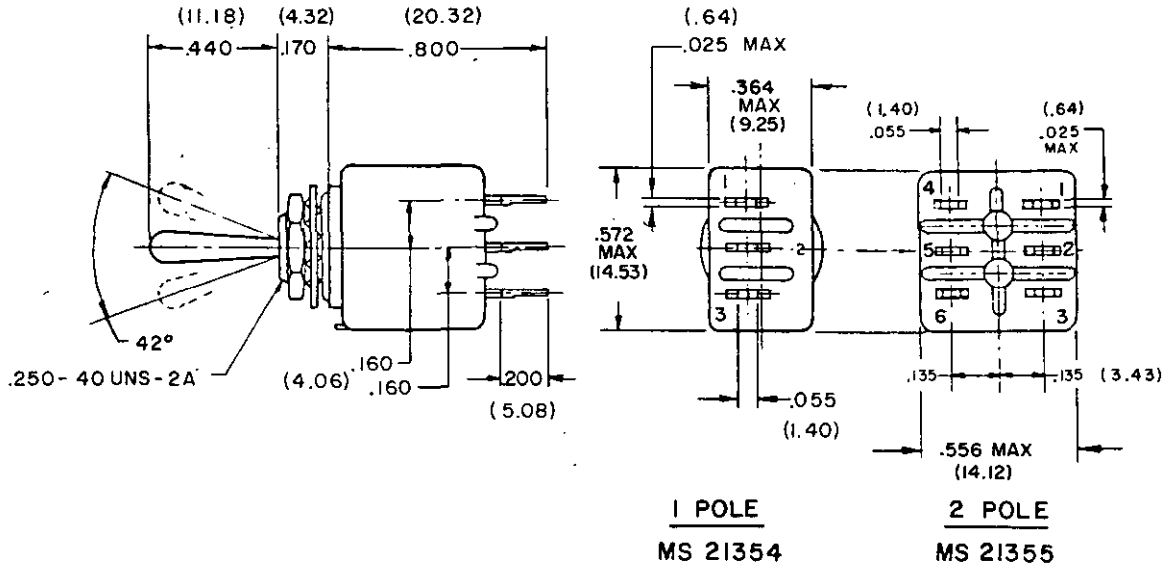
1/ 800 series have silicone bonded seals (for MS21353 only); 300 series have mechanical seals.
2/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

FIGURE 112-19. MS21352 and MS21353 toggle switches, positive break, special circuit, miniature, toggle sealed, solder lug, double pole (MIL-S-8834).

Supersedes page 112.21
of 20 June 1977

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Dash number	1/ Circuit with toggle lever in			Current capacity (amperes)						Life low current level switching 5 mV
				Resistive load			Inductive load			
	Opposite locking tab side	Center	Locking tab side	28 Vdc	115 Vac 60 Hz	115 Vac 400 Hz	28 Vdc	115 Vac 60 Hz	115 Vac 400 Hz	
										(μA)
-211	ON	OFF	ON							
-221	ON	NONE	OFF							
-231	ON	NONE	ON							
-241	ON	OFF	NONE	5	2	3	1	1	2	25
-271	MOM-ON	OFF	MOM-ON							
-281	NONE	OFF	MOM-ON							
-311	ON	OFF	MOM-ON							

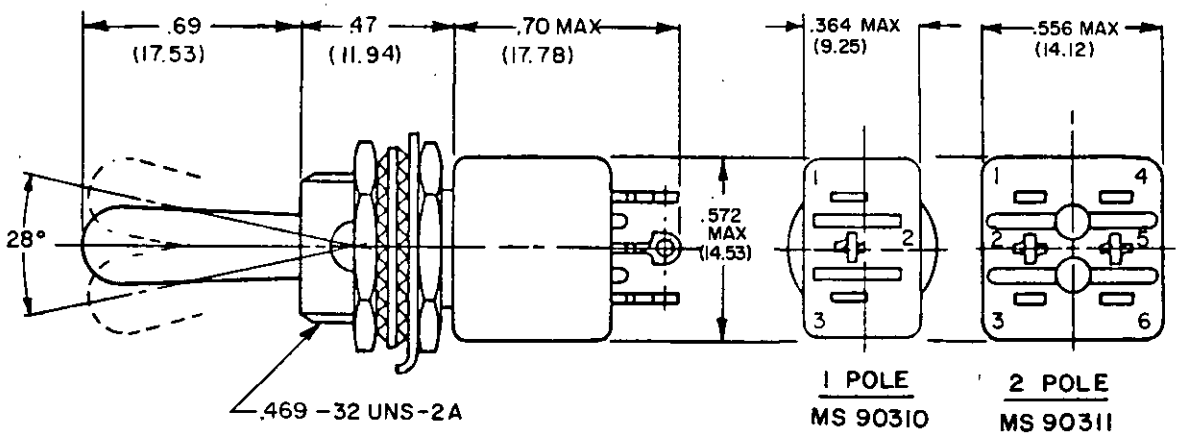
1/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

FIGURE 112-20. MS21354 and MS21355 toggle switches, positive break, miniature, toggle sealed, printed circuit board terminals, .250 mounting bushing (MIL-S-8834).

Supersedes page 112.22
of 23 March 1976

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MS dash number 1/		2/ Circuit with toggle lever in			Current capacity (amperes)						Life low current level switching 5 mV
					Resistive load			Inductive load			
		Opposite keyway side	Center	Keyway side	28 Vdc	115 Vac		28 Vdc	115 Vac		
						60 Hz	400 Hz		60 Hz	400 Hz	
											(μA)
-711	-211	ON	OFF	ON							
-721	-221	ON	NONE	OFF							
-731	-231	ON	NONE	ON							
-741	-241	ON	OFF	NONE	5	2	3	1	1	2	25
-771	-271	MOM-ON	OFF	MOM-ON							
-781	-281	NONE	OFF	MOM-ON							
-811	-311	ON	OFF	MOM-ON							

1/ 700 and 800 series have silicone bonded seals (applicable to MS90311 only); 200 and 300 series have mechanical seals.
2/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

FIGURE 112-25. MS90310 and MS90311 toggle switches, miniature, toggle sealed, solder lug, .469 mounting bushing (MIL-S-8834).

Supersedes page 112.27
of 20 June 1977

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LOAD	ELECTRICAL RATINGS		
	Sea level		50,000 feet
	28 Vdc	115 Vac, 60 Hz	28 Vdc
	amperes	amperes	amperes
Resistive - - -	5	5	5
Inductive - - -	3	5	2.5
Lamp- - - - - -	2.4	1.5	2.4

INCHES	MM
.060	1.52
.115	2.92
.36	9.14
.470	11.94
.500	12.70
.720	18.29
.860	21.84
.990	25.15
1.125	28.58
1.160	29.46

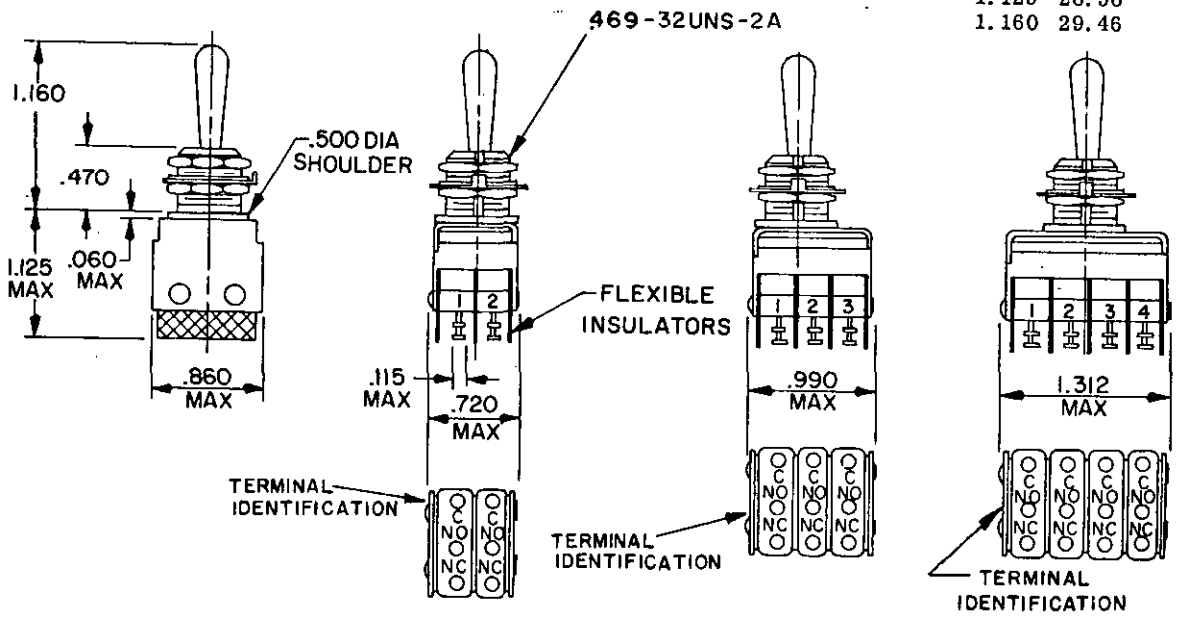
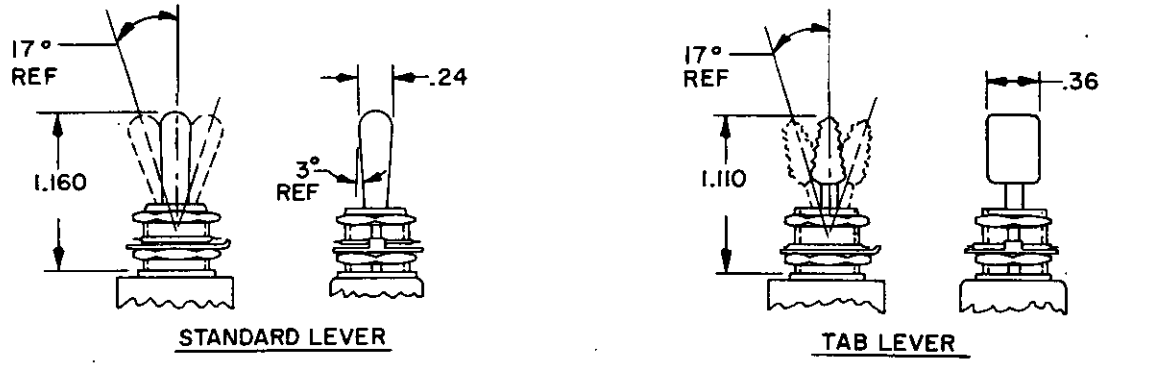


FIGURE A FIGURE B FIGURE C



STANDARD LEVER TAB LEVER

FIGURE 112-26. Switch assemblies, sensitive, toggle lever (with integral leaf), 5 amperes, unsealed (MIL-S-8805/26).

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TABLE I. Part numbers and characteristics.

Dash No.	Figure	Locking combination	Circuit configuration (see table II)	Switch action		
				Opposite keyway	Center	Keyway
-001	A	D	1	Main	None	Main
-002	C	D	4	Main	None	Main
-003	C	F	4	Main	None	Main
-004	C	G	4	Main	None	Main
-005	A	E	2	Mom	Main	Mom
-006	A	A	2	Main	Main	Main
-007	A	D	2	Main	Main	Main
-008	A	E	2	Main	Main	Main
-009	A	F	2	Main	Main	Main
-010	A	G	2	Main	Main	Main
-011	A	J	2	Main	Main	Main
-012	A	K	2	Main	Main	Main
-013	A	L	2	Main	Main	Main
-014	A	M	2	Main	Main	Main
-015	A	N	2	Main	Main	Main
-016	A	B	2	Mom	Main	Main
-017	A	E	2	Mom	Main	Main
-018	A	P	2	Mom	Main	Main
-019	B	A	3	Main	Main	Main

TABLE II. Circuit configuration.

Circuit configuration	Switch pole number	Circuit closed with toggle lever in		
		Opposite keyway side	Center	Keyway side
1	1	C-NC	None	C-NO
1	2	C-NC	None	C-NO
2	1	C-NO	C-NC	C-NC
2	2	C-NC	C-NC	C-NO
3	1	C-NO	C-NC	C-NC
3	2	C-NC	C-NC	C-NO
3	3	C-NO	C-NC	C-NC
4	1	C-NC	None	C-NO
4	2	C-NC	None	C-NO
4	3	C-NC	None	C-NO
4	4	C-NC	None	C-NO

OPERATING CHARACTERISTICS:
Coincidence of operating and releasing points: All poles transfer within 10 degrees of lever travel.
Locking arrangement: Positive locking is accomplished and prevents motion of the toggle lever until the locking mechanism is manually released.
Locking mechanism release: The force required to release the locking mechanism is 3 to 5 pounds.

PART NUMBER: Consists of "M8805/92" followed by the dash number shown in table I.

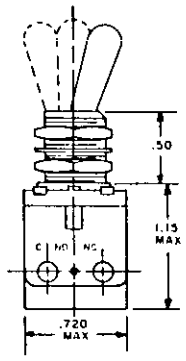
FIGURE 112-27. Switch assemblies, sensitive, lever lock, 5 amperes, unsealed (MIL-S-8805/92) - Continued.

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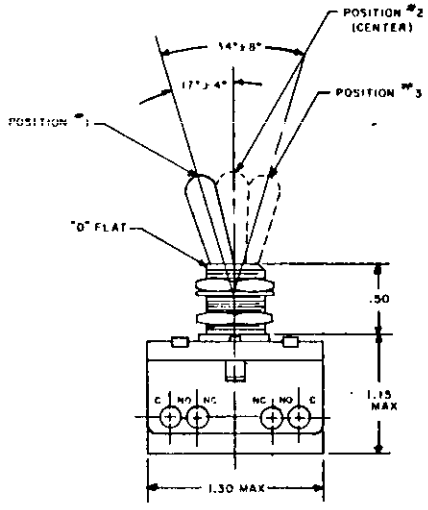
LOAD	ELECTRICAL RATINGS (amperes)		
	Sea level		50,000 feet
	28 Vdc	115 Vac, 60 Hz	28 Vdc
Resistive - - - -	7.0	7.0	4.0
Inductive - - - -	4.0	7.0	2.5
Lamp- - - - - - -	2.5	2.0	2.5

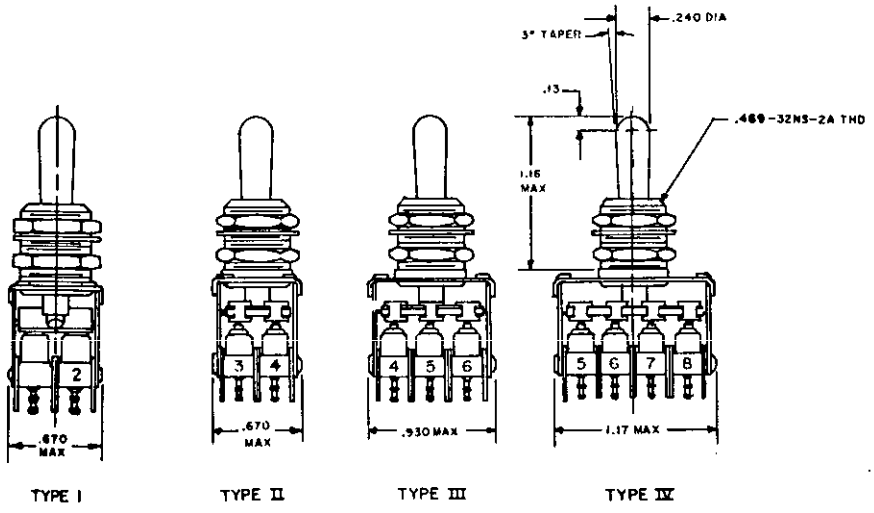
INCHES	MM
.13	3.30
.240	6.10
.50	12.70
.670	17.02
.720	18.29
.930	23.62
1.15	29.21
1.16	29.46
1.17	29.72
1.30	33.02

SIDE VIEW FOR TYPE I



SIDE VIEW FOR TYPES II THROUGH IV





TYPE I TYPE II TYPE III TYPE IV

Part No. M8805/93-				Operation
Type I 2PDT	Type II 4PDT	Type III 6PDT	Type IV 8PDT	
001	008	015	022	Maint-maint-maint
002	009	016	023	Mom-maint-maint
003	010	017	024	Maint-maint-mom
004	011	018	025	Mom-maint-mom
005	012	019	026	Maint-none-maint
006	013	020	027	Mom-none-maint
007	014	021	028	Maint-none-mom

FIGURE 112-28. Switch assemblies, sensitive, lever, unsealed, 7 amperes (MIL-S-8805/93).

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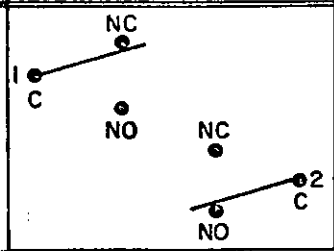
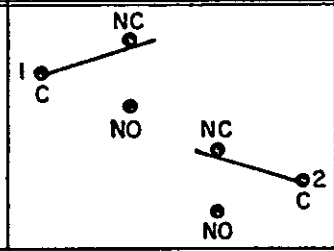
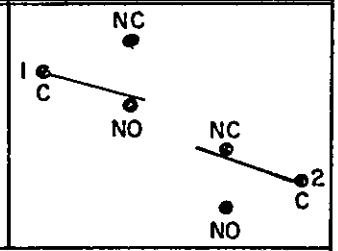
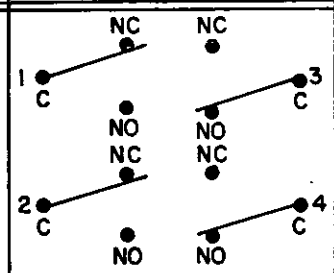
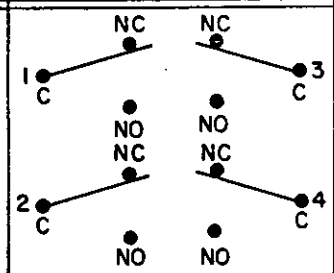
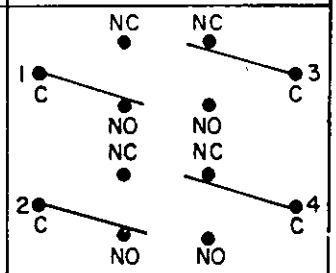
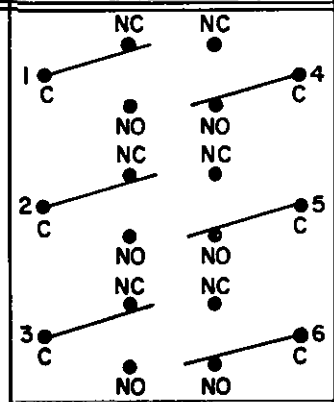
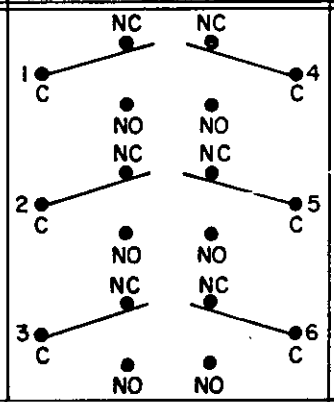
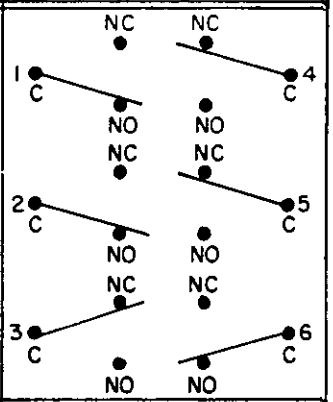
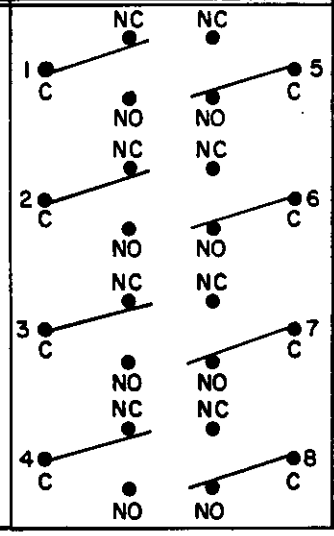
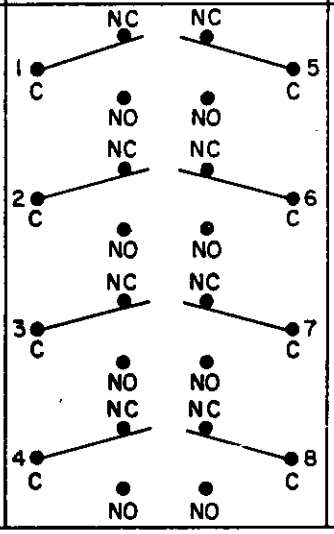
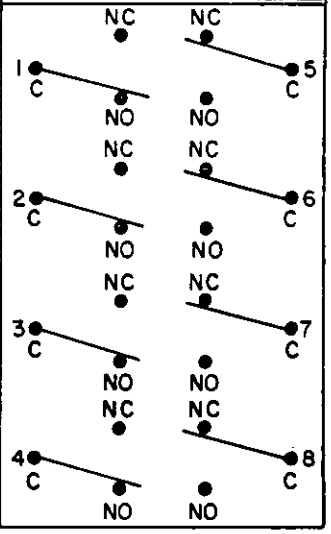
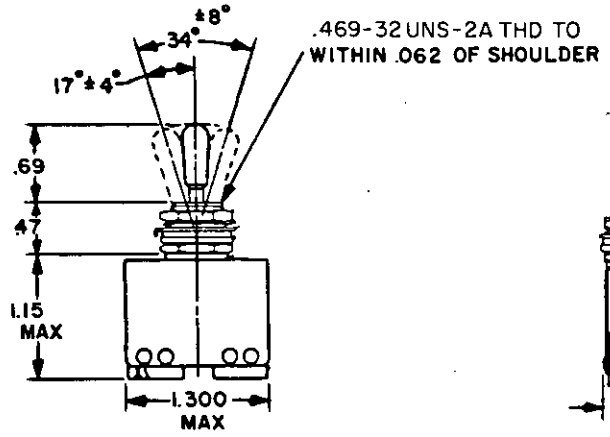
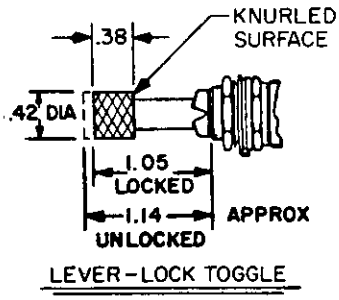
TYPE	SCHEMATIC DIAGRAM		
	POSITION NO.1 ("D" FLAT)	POSITION NO.2 (CENTER)	POSITION NO.3 (OPPOSITE)
I			
II			
III			
IV			

FIGURE 112-28. Switch assemblies, sensitive, lever, unsealed, 7 amperes (MIL-S-8805/93) - Continued.

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LOAD	ELECTRICAL RATINGS (amperes)			
	Silver contacts			Gold contacts
	Sea level		70,000 ft	Sea level and 70,000 ft
	28 Vdc	115 Vac, 60 Hz	28 Vdc	28 Vdc
Resistive - - -	7.0	7.0	3.0	1.0
Inductive - - -	4.0	7.0	2.0	.5
Lamp- - - - - -	2.5	2.0	---	---





LEVER-LOCK TOGGLE

INCHES	MM
.062	1.57
.38	9.7
.42	10.7
.47	11.9
.670	17.02
.69	17.5
.930	23.62
1.05	26.7
1.14	29.0
1.15	29.2
1.30	33.0
1.170	29.72

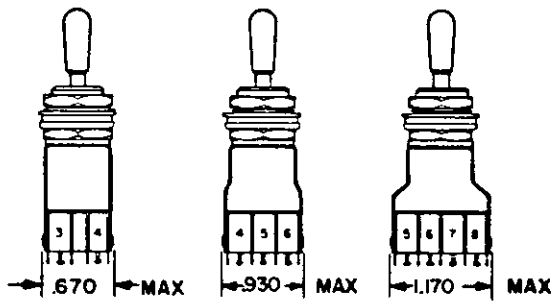


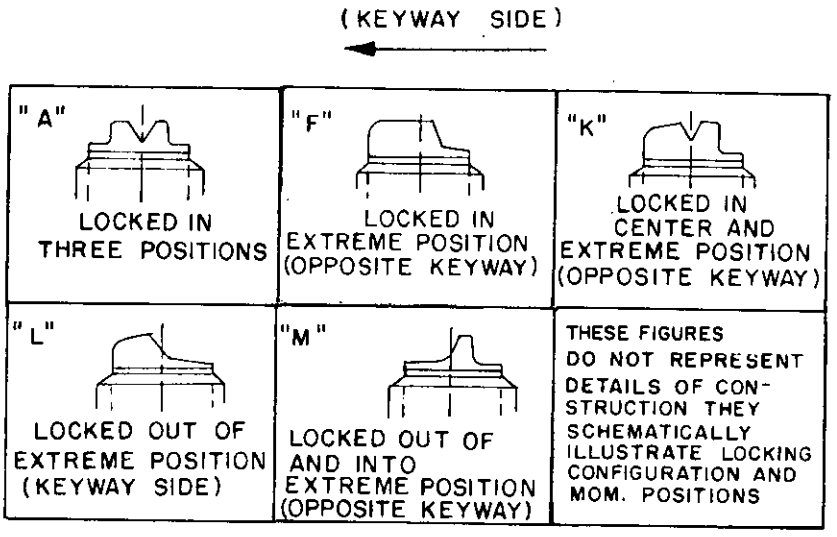
FIGURE A (4-POLE) FIGURE B (6-POLE) FIGURE C (8 POLE)

BASIC SWITCH AND TERMINAL IDENTIFICATION

Type number	Basic switch numbers			With toggle lever in position					
				Keyway side		Center		Opposite keyway side	
	Figure A	Figure B	Figure C	Toggle action	Circuit closed	Toggle action	Circuit closed	Toggle action	Circuit closed
1	1-2 3-4	1-2-3 4-5-6	1-2-3-4 5-6-7-8	Maintain	C-NC C-NO	Maintain	C-NC C-NC	Maintain	C-NO C-NC
3	1-2 3-4	1-2-3 4-5-6	1-2-3-4 5-6-7-8	Maintain	C-NC C-NO	(None)		Maintain	C-NO C-NC
5	1-2 3-4	1-2-3 4-5-6	1-2-3-4 5-6-7-8	Momentary	C-NC C-NO	Maintain	C-NC C-NC	Maintain	C-NO C-NC

FIGURE 112-29. Switch assemblies, sensitive, toggle actuated, 7 amperes, panel sealed, RFI grounded (MIL-S-8805/103).

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Part number M8805/103			Type number	Locking combination	Contact material
Figure A (4-Pole)	Figure B (6-Pole)	Figure C (8-Pole)			
-001	---	-002	1	None	Silver
---	-004	---	3	None	
-005	---	---	5	None	
-009	---	---	5	M	
---	-010	---	3	F	
---	---	-011	1	A	
-012	---	-013	1	None	Gold
-014	---	-015	5	None	
-016	---	---	5	K	
-017	---	---	1	M	
---	---	-018	5	L	

FIGURE 112-29. Switch assemblies, sensitive, toggle actuated, 7 amperes, panel sealed, RFI grounded (MIL-S-8805/103) - Continued.

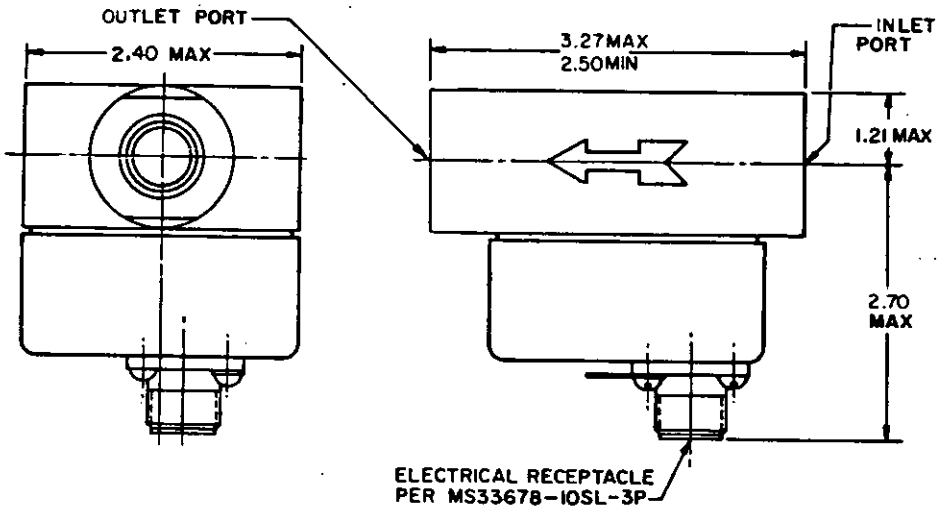
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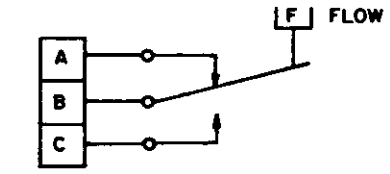
SECTION 114

SWITCHES, AIR AND LIQUID FLOW
(Applicable specification: MIL-S-28788)

FIGURE AND SPEC SHEET NO.	SWITCH CHARACTERISTICS				PAGE
	Electrical rating at 28 Vdc (amperes)	Operating temperature	Proof pressure	Burst pressure	
114-1 - MIL-S-28788/1	3 inductive	32° to 185°F	150 psig	300 psig	114.2
114-2 - MIL-S-28788/2	3 inductive	0° to 65°C	150 psig	300 psig	114.3

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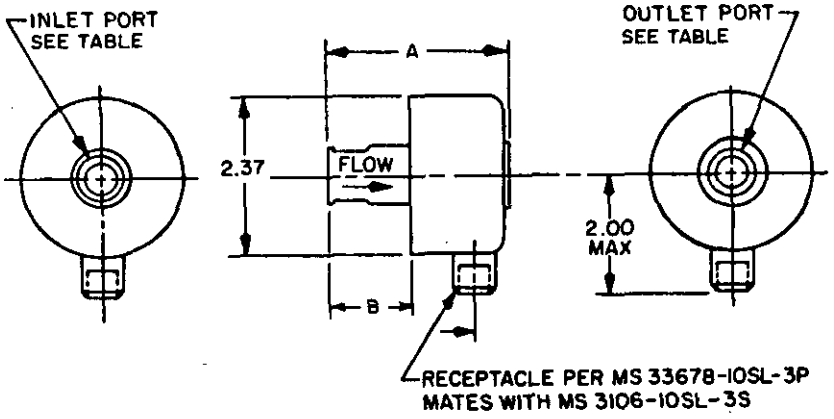
WIRING DIAGRAM
(SHOWN WITH ZERO FLOW)

INCHES	MM
1.21	30.7
2.70	68.6
2.40	61.0
2.50	63.5
3.27	83.1

Military part number	Port size per MS16142		Switch flow rates		Pressure drop during flow	
	Inlet	Outlet	Actuation increasing GPM (max) at 104° ±9°F	Deactuation decreasing GPM (min) at 104° ±9°F	PSIG (max)	Rated flow GPM
M28788/1-1	-6	-6	.85	.60	5	1.00
-2	-6	-6	.42	.27	5	.50
-3	-6	-6	.60	.40	4.5	.66
-4	-6	-6	.65	.45	5	.75
-5	-8	-8	3.70	2.90	5	4.0
-6	-6	-6	1.30	1.00	5	1.50
-7	-6	-6	1.80	1.50	5	2.00
-8	-6	-6	2.30	2.00	5	2.50
-9	-8	-8	2.70	2.40	5	3.00
-10	-8	-8	3.70	3.20	7	4.00
-11	-8	-8	4.50	4.00	7	5.00
-12	-8	-8	5.40	4.80	7	6.00

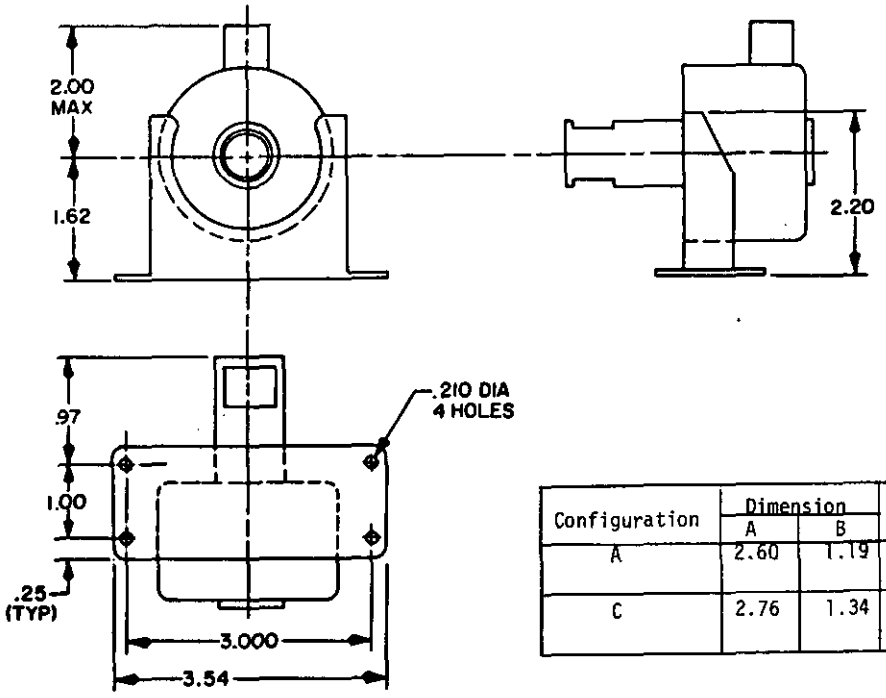
FIGURE 114-1. Switches, liquid flow (MIL-S-28788/1).

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CONFIGURATION A AND C

INCHES	MM
.210	5.33
.25	6.4
.97	24.6
1.00	25.4
1.62	41.1
2.00	50.8
2.20	55.9
2.37	60.2
3.000	76.2
3.54	89.9

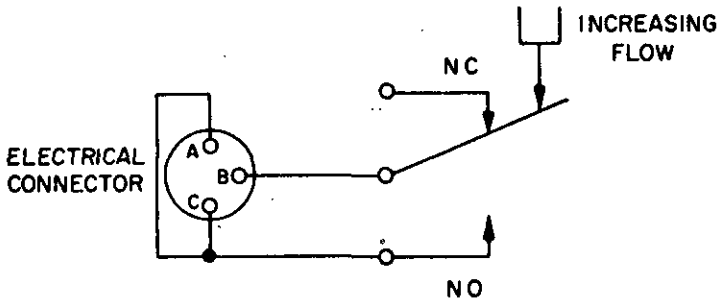


CONFIGURATION B

Configuration	Dimension		Port size
	A	B	
A	2.60	1.19	MS33649-6 or AND10050-6
C	2.76	1.34	MS33649-8 or AND10050-8

FIGURE 114-2. Switches, liquid flow (MIL-S-28788/2).

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SCHEMATIC DIAGRAM
(ALL CONFIGURATIONS)

Part number M28788/2-	Configuration	Actuating flow increasing GPM max	Deactuating flow decreasing GPM min	Maximum allowable pressure drop	Fluid media
01	A	0.45	0.3	4 PSI at 0.3 GPM	I
02	A	0.65	0.5	5 PSI at 0.5 GPM	I
03	A	0.85	0.7	6 PSI at 0.7 GPM	I
04	A	1.40	1.2	4 PSI at 1.2 GPM	I
05	A	3.25	3.0	4 PSI at 3.0 GPM	I
06	B	4.50	4.0	5 PSI at 4.0 GPM	II
07	A	0.55	0.45	5 PSI at 0.5 GPM	II
08	C	3.10	2.7	5 PSI at 3.0 GPM	II
09	C	3.00	2.5	5 PSI at 2.5 GPM	II
10	C	2.80	2.3	5 PSI at 2.3 GPM	II
11	C	10.0	7.5	5 PSI at 7.5 GPM	II

FIGURE 114-2. Switches, liquid flow (MIL-S-28788/2) - Continued.

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